

# Baseline Report

DSE Group 28

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# 1. Executive Overview

Because of the aviation industry's move towards net-zero global emissions in the coming decades, optimisation of aircraft configurations and performance improvements have been the focus of aircraft designers in recent years. These improvements can be achieved through implementing more lightweight structures of lower stiffness, but also through certain aerodynamic features, such as increasing the wing's aspect ratio, the use of winglets and folding wing-tips. As a consequence, new challenges arise in the form of aeroservoelastic interactions between airflow, structures, rigid-body motions, and control systems.

The aforementioned phenomena can be static or dynamic, some of the former being torsional divergence and control reversal, and some of the latter being flutter. Such effects can occur on any part of the airplane, such as the main wing, the empennage or the control surfaces. In general, they can lead to catastrophic failure and have to be avoided within the operational flight envelope. This is often achieved with the use of aeroelastic tailoring; a passive approach where the structural elements (such as composite panels) are stiffened in a specific direction, or mass-balances which are attached to the wing to modify the inertial properties. While these passive approaches do mitigate the problems to a certain extent, they optimize the airplane for a specific flight condition, and often come at the cost of additional weight.

An alternative, emerging approach, is to design active control systems which prevent the onset of possibly catastrophic oscillations (active flutter suppression systems), actively alleviate loads (manoeuvre and gust load alleviation systems) and improve passenger-comfort [1]. These systems rely on robust and adaptive control algorithms, the feedback and feed-forward information from sensors, as well as control surfaces powered by high-bandwidth actuators that physically apply the corrective control inputs. The benefit of these active approaches is not only higher efficiency in terms of better performance, handling characteristics, and increase in safety, but also in terms of expanding the operational flight envelope.

Nevertheless, despite these advantages, the testing and certification of these flight control systems remain a challenge. Accurate computational simulation is difficult due to the inherent nonlinearity of high-aspect-ratio wings. On the other hand, wind tunnel tests cannot accurately model the real behaviour of such aircraft both due to scale and because the model is constrained in the wind tunnel. This is important as rigid-body motions are a crucial component for the excitation of phenomena such as body freedom flutter. Lastly, wind tunnel testing of flexible aircraft models is not only extremely expensive, but it is sometimes also prohibited in specific situations due to the risk of damaging the wind tunnel. For instance, in case of testing the post-flutter behaviour [2]. The remaining approach is flight testing, an approach that provides realistic data, but notoriously precarious for its cost and pilot safety.

Therefore, the aim of the High-speed Unmanned General-purpose Observatory (HUGO) project, is to design a high-aspect-ratio turbine-powered flying test-bed which is low-cost, low-weight, and versatile. It must assist researchers in testing aeroservoelastic control systems, by having the ability to carry multiple types of scientific payloads. Consequently, the idea is to create a versatile design where the test-bed itself can be modified to be able to carry the payload. This will involve the design and analysis of modular structural and aerodynamic features, the integration of the provided VULCAN engine for reaching near-transonic speeds, performance analysis, ground operation analysis, manufacturing procedure and an analysis of environmental impact.

Similar scaled aircraft for testing aeroelastic effects and control laws have been developed before. For example, in 2005, Lockheed Martin's Skunkworks began the body freedom flutter research program, leading to the development of the X-56A, with the goal of investigating active flutter suppression, and gust load alleviation of high aspect ratio wings <sup>(1)</sup>. The NLR has also developed the scaled flight demonstrator (SFD); a scale model of the A320, aiming to supplement numerical simulations, wind tunnels, and other experimental testing means, by investigating aircraft dynamic behaviour, and then pivoting to distributed electric propulsion <sup>(2)</sup>. Although similar to HUGO, the X-56A, and SFD are both highly specialised in their implementation. Even though the

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<sup>1</sup>[Lockheed Martin X-56A](#)(accessed on 29.04.2026))

<sup>2</sup>[NLR Scaled Flight Demonstrator Maiden Flight](#)(accessed on 29.04.2026))

X-56A includes a modular wing, its main purpose is not to serve as a test bed for versatile experimentation. Additionally, neither aircraft flies above 150kts (277.8 km/h), staying well within the incompressible flow regime (2) (3). HUGO provides a significant advantage as it goes faster (in the transonic regime), and would allow for testing a much wider range of experiments, aiming to replace wind tunnel and flight testing within its niche.

Based on the user requirements and the existing configurations, the project objectives have been defined, comprising financial, quality, performance, business, and compliance objectives. Financial objectives relate to the maximum manufacturing cost per configuration of 45000€ and the maximum operating cost of 2000€. Quality objectives ensure that the scientific measurements can be collected with satisfactory standards. Performance objectives relate to maximum speed, endurance and payload capacity. Business objectives specify the required profit and market share margins. Finally, compliance objectives relate to compliance with airworthiness requirements.

Using the objectives, the market has been analysed to obtain a clear overview of the possible customers of such flying laboratories, as well as the strengths, weaknesses, opportunities, and threats (SWOT). Universities and research institutes, as well as aerospace companies and defence agencies, have been identified as potentially interesting market segments, with the first two having been identified as the most relevant for this project. Additionally, the overcoming of the limitations of wind tunnels, and market threats stemming from flight certification have been identified as the key opportunities and threats, respectively.

Furthermore, the functional analysis has been performed based on the project objectives and market analysis. That entailed the mission profile definition, and creation of the functional flow diagram and functional breakdown structure diagrams. Therein, the mission profile has been subdivided into the launch, flight test, and recovery phases. Moreover, the requirement discovery tree has been created and the final list of system requirements has been compiled. In this final list, the key and driving requirements have been indicated as well.

The design process is constrained by the available technology and by stakeholder requirements. These constraints define limits with which the system has to be developed. Following an analysis, the most important limitations are brought by the centre of gravity location, mass, communication link capabilities, aircraft dimensions, quantity of fuel, available power and human power. Each of these resources must be quantified and monitored to ensure that HUGO remains feasible throughout the design process. To reduce the risk of resource misallocation and inefficient design decisions, a technical budget was established. This budget provides a structured overview of the main resources and identifies the key factors influencing each allocation. At the moment, due to the early stage that the project resides into, a quantitative approach for determining the budget cannot be applied for each resource, and only a preliminary distribution was created. Effective tracking of the budgets will take place by centralising all the available information and frequently updating the resource states.

In order to select feasible configurations for further design and analysis, a design option tree has been created, listing all reasonable concepts the design team could think of. Among other, this included different wing and tail configurations, fuel tank architecture, engine mounting locations and number of engines, and undercarriage configurations. These numerous options have been reduced by eliminating the clearly unfeasible options, followed by the selection of the most feasible options among the remaining ones. That yielded the following four configurations:

1. Low wing, T-tail, retractable gear: the traditional option, mimicking standard commercial aircraft configurations
2. Low wing, H-tail, retractable gear: offers additional possibilities for engine placement and tail-wake interference

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<sup>3</sup> [Military Factory Lockheed Martin X-56A](#) (accessed on 29.04.2026))

3. Low wing, inverted V-tail, retractable gear: offers additional possibilities for test specimen mounting, and cleaner aerodynamic flow over the top of the aircraft
4. Flying wing with buried engine(s): low parasitic drag, simulation of coupled aeroelastic phenomena such as the Body Freedom Flutter

In the baseline sustainability assessment, the aspects of construction, operation and end-of-life solutions were considered. The conclusions were that construction and manufacturing should not only be evaluated based on productions emissions but also for the potential reduction in emission arising in a lower weight. For the operational aspects it was noted that the use of SAF (sustainable aviation fuel) as fuel might mitigate the increase of carbon-emissions in the atmosphere but the production of NOX would still be a factor. It was also found that the end of life solutions would be heavily influenced by certain design choices regarding bonding and joining materials together when creating the aircraft. Finally lifecycle assessment and streamlined lifecycle assessment were introduced as tools to formalize the environmental impact of different design configurations.

The profitability and return analysis determined the size of the markets for aerospace component testing, the total cost of goods sold and the profitability of HUGO, resulting in a total attainable market share in the Netherlands for a target year. The total addressable global market was found to be \$6.94B in 2026, and \$1B in Europe. Data for the serviceable addressable market in the Netherlands was not available, so it was approximated using a bottom-up approach, resulting in €71M, and in a final serviceable obtainable market of €14.2M. As seen in [Table 9.1](#), the production and direct operational costs; estimated as a an expense per flight day or per 300 flight days, are listed. These were used to calculate the cost of goods sold (COGS) per flight day of €1142 at a price of €2000. For a total of 100 flight days per year, this represents a net annual profit of €63,664 per HUGO.

*Table 9.1: Overview of UAV costs and their contribution to flight day pricing*

| Type                                                    | Subcategory  | Description                          | Pricing Role | Cost (€) |
|---------------------------------------------------------|--------------|--------------------------------------|--------------|----------|
| <b>Direct costs, reference flight day</b>               |              |                                      |              |          |
| Operations                                              | Turbine fuel | 11.5 L of SAF, 4 test flights        | Direct       | 120      |
| Operations                                              | Electricity  | Batteries, computers, systems        | Direct       | 80       |
| Operations                                              | Transport    | Vehicle + logistics                  | Direct       | 250      |
| Operations                                              | Mission      | UAV pilot/operator                   | Direct       | 400      |
| <b>Fractional costs, full return in 300 flight days</b> |              |                                      |              |          |
| Certification                                           | RDW + ILT    | Registration and flight permits      | Fraction     | 5 000    |
| Production                                              | Materials    | UAV structural and system components | Fraction     | 28 000   |
| Production                                              | Labour       | 300 h at 50 €/h                      | Fraction     | 15 000   |
| Production                                              | Facility     | 10 days at 200 €/day                 | Fraction     | 2 000    |
| Operations                                              | Insurance    | 500 € annually, 15 years total       | Fraction     | 7 500    |
| Maintenance                                             | Engine       | 250 € per 50 h, 3000 h total         | Fraction     | 15 000   |
| Maintenance                                             | Airframe     | 500 € per 100 h, 3000 h total        | Fraction     | 15 000   |

The primary assumptions for the profitability analysis were identified as the percentage of SAM that can be serviced by HUGO, the certification cost, and the magnitude of the maintenance costs. Future work shall conduct a sensitivity study on those three parameters to quantify their impact on overall financial viability of the project. The optimum time for this study to be conducted is when maintenance costs are estimated in more detail for different aircraft configuration considered for trade-off.

The technical risk assessment identified several development and operational risks, and ranked them according to likelihood and severity. The most likely and severe development risks were found to be discrepancies in component size and weight specified by the manufacturer vs actual values, system model errors (inadequate

V&V). The most likely and severe operational risks were mechanical failure of components, exceeding flight limit area, and sudden component detachment during normal operations. Subsequently, a preliminary specific operation risk assessment (SORA) was performed. SORA is a 10 step classification methodology used for determining the operation risk of a UAV placing it in a specific assurance and integrity level (SAIL) category. As HUGO has not yet been developed, and a flight location has not been determined, it is not possible to make a SAIL classification at this time. However, if HUGO were to operate within controlled airspace, the preliminary classification was found to be SAIL II. However, for non-controlled airspace with a population density greater than 5 people per square kilometre, the classification was predicted to become SAIL V or SAIL VI, requiring EASA certification, incurring higher development costs.

In conclusion, the limitations of conventional wind tunnels and flying laboratories for performing aeroservoelastic experiments have been identified as the main incentives for the HUGO project. While similar laboratories do exist, such as X-56A and SFD, they are limited in terms of versatility, maximum speed and cost-efficiency. Subsequently, project objectives, market, functional and requirement analyses, and resource allocation have been performed, followed by the selection of the four best-performing configurations. Those will serve as the starting point for the mid-term phase of the project. Last but not least, the sustainability, profitability and technical risk analyses have been performed.

## 2. Project Objectives

Project objectives should be specific, measurable, actionable, relevant, and time-bound (the SMART model) outcomes that a project intends to deliver. This project aims at designing, testing and marketing an unmanned aero vehicle specialised in testing aeroelastic phenomena and mitigation strategies, providing the scientific community with a cheaper and more accurate alternative to modern subsonic wind tunnels.

### 2.1. Financial

The financial objectives focus on the budget management and profitability of the project.

1. The total manufacturing cost per aircraft configuration shall not exceed €45000, including materials, propulsion system, avionics, and all other aircraft components [2].
2. The aircraft shall have a maximum operational cost of €2000 per flight day, including all consumables, operations, and maintenance (chapter 9).
3. The aircraft shall have an engine maintenance interval of maximum 50 hours (subsection 9.5.4) to ensure cost-effective lifecycle performance.
4. Components shall be replaced instead of repaired when the repair cost exceeds 30% of the replacement cost, ensuring economically viable maintenance <sup>(4)</sup>.

### 2.2. Quality

The quality objectives are used to define standards for the deliverables.

1. Repeated measurements under identical atmospheric conditions shall not deviate by more than  $\pm 1\%$ , consistent with typical air data measurement accuracy in practice.
2. The aircraft shall achieve a minimum of 3000 flight hours between critical failures (MTBF) (chapter 9).
3. Data shall be stored in a standardized format compatible with analysis tools.
4. The aircraft shall ensure collection and storage of all payload sensor data during flight operations, including discrete, analog, and digital signals <sup>(5)</sup>.
5. The aircraft shall provide mounting integration of external sensors without requiring modification of core flight control systems.

### 2.3. Performance

The performance objectives focus on the results and effectiveness of the project.

1. The aircraft shall enable continuous flight and collection of flight test data at 0.75 Mach [2] for at least 5 minutes at an altitude of TBD km per mission (*the altitude will be determined from certification requirements and the flight regime*).
2. The aircraft shall enable the integration and operation of payloads of at least 5 kg and 5 L to support its intended flight test and data acquisition missions [2].
3. The aircraft configuration shall allow execution of at least 4 repeatable flight tests per test day (chapter 9).
4. The UAV shall demonstrate a minimum operational range of TBD km using a constrained flight test profile (out-and-back) within regulated airspace (*the range depends on the airspace where the tests will be performed*).

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<sup>4</sup>Towards a threshold for “reasonable price” of spare parts (accessed: 28 April 2026).

<sup>5</sup>Flight Data Acquisition Unit (FDAU) (accessed: 30 April 2026)

## 2.4. Business

The business objectives align with strategic goals like increased revenue or market share.

1. The project shall achieve a cumulative net financial profit of approximately €64K per HUGO, per year after initial operational deployment, based on total development, costs of goods sold, and operational costs relative to generated revenue ([chapter 9](#)).
2. The project shall achieve a yearly market share of 4.5% within the Dutch target market of universities, research centres, and aircraft companies after initial operational deployment, measured based on the deployment of 10 HUGO, each with 100 contracted testing days per year representing a total net profit of €636K after tax <sup>(6)</sup>.
3. HUGO shall be ready for initial operational deployment within 10 working days from the production process start ([section 9.4](#)).

## 2.5. Compliance

The compliance objectives ensure adherence to regulations and standards.

1. The UAV system shall demonstrate compliance with applicable airworthiness and safety requirements for unmanned aircraft, including fail-safe and emergency procedures, prior to deployment.
2. All communication systems shall comply with applicable radio frequency regulations and spectrum usage requirements within the region of operation.
3. The UAV system shall comply with applicable environmental regulations, including noise and emissions limits, within the intended operational environment.

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<sup>6</sup>[Corporate income tax-Government of Netherlands](#)(accessed: 24 April 2026)

## 3. Market analysis

The main goal of a market analysis is to research and establish the state of the competition by several criteria, including cost, production capacity or logistics and reach.

### 3.1. Market segmentation

Market segmentation is the process of dividing a broad market into smaller and concrete groups with similar interests. [3] It is crucial for defining a strong business marketing strategy. This helps to understand and meet current market requirements as well as distinguishing from the competition. Additionally, a beachhead market needs to be identified. The beachhead market is the initial and most promising market aimed for the product to enter first. An overview of the main identified market segments is shown in [Table 3.1](#).

#### Universities and academic laboratories

Universities are the most accessible market segment and, therefore, the beachhead market. Universities typically conduct research in aeroelasticity, flight dynamics, active control and other related topics, but often lack access to large wind tunnels or manned flying laboratories. Therefore, a fleet of testbed UAVs at a fraction of the cost of a wind tunnel could be utilized in their research instead. As an example, the University of Washington is aiming to research similar phenomena, which makes it a suitable potential customer. (7)

#### Research centres

A second key segment consists of national aerospace research centres and applied research institutes, such as NLR or DLR. These institutions conduct research in multiple related fields, including aeroelasticity or flutter suppression. Compared with universities, they usually have stronger technical resources and larger budgets. Their main interest is generating high-quality and accurate experimental data. Robustness and predictability of the modular platform is of high importance for these customers. (8)

#### Aerospace companies

Companies developing new aircraft and subsystems for aircraft follow as other potential customers. These stakeholders may make use of the platform to test new aerospace components in real-life conditions prior to certification. Since the testing often involves many design iterations and testing loops, the costs of the process can grow quickly. At the same time, the use of external facilities for testing can delay the process further. Therefore, shortening development cycles allows components and configuration to enter market earlier would save costs and resources for the aerospace manufacturers. These companies usually allocate a budget for research and development (R&D) activities, which can help to make this project profitable.

#### Defence sector

Recently, governments has started allocating more resources to defence and security (9). Defence companies and governmental departments need to test new technologies to be used in the rapidly evolving sector of unmanned aerial vehicles (UAVs) for combat and reconnaissance. Having a dedicated test platform could help these organizations centralize the testing of new and sensitive technologies instead of relying on an external partner.

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<sup>7</sup> [Active Aeroelastic Structures Lab, University of Washington](#) (accessed on 29.04.2026)

<sup>8</sup> [About DLR](#) (accessed on 29.04.2026)

<sup>9</sup> [EU defence in numbers](#) (accessed 29.04.2026)

**Table 3.1:** *Market segmentation overview*

| Segment                      | Typical users                              | Main need                                                                     | Attractiveness |
|------------------------------|--------------------------------------------|-------------------------------------------------------------------------------|----------------|
| Universities                 | TU Delft and other universities/faculties  | Affordable modular platform for research, teaching, and student projects      | High           |
| Research centres             | NLR, DLR, national labs                    | High accuracy validation of aeroelastic and flight control technologies       | High           |
| Aircraft companies           | Aerodynamics and structure engineers       | Accelerate testing and development cycles, reduce dependency                  | Medium to High |
| Defense and government users | Government labs, defense research agencies | Centralize testing and development, reduce dependency, accelerate development | Medium         |

### 3.2. SWOT analysis

A SWOT analysis evaluates a project or organisation by examining its strengths, weaknesses, opportunities, and threats, helping to inform strategic decision-making, highlight key risks, and guide future planning. An organisation can use a SWOT analysis to assess both internal and external factors, enabling it to develop strategies that leverage strengths, mitigate weaknesses, and take advantage of emerging opportunities. It also provides a structured framework for evaluating competitive advantage and informing future strategic direction.

The classification groups strengths and opportunities, and weaknesses and threats.

**Table 3.2:** *Strengths and Opportunities of the Aircraft*

| Strengths                                                                                                                                                                                            | Opportunities                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Enables data collection in unconstrained flow, avoiding wind tunnel artefacts such as wall effects and mounting interference [2].                                                                    | Testing to failure, including structural limits, flutter, and failure modes that are impractical or unsafe to investigate in traditional wind tunnel facilities [2]. |
| Eliminates or reduces scaling issues (e.g. Reynolds number mismatch), improving fidelity of aerodynamic and aeroelastic data [2].                                                                    | Limitations of wind tunnel testing, such as wall interference, scaling issues, and cost create a gap that the modular flying test bed can fill [2].                  |
| Realistic operation conditions that are hard to simulate in wind tunnel facilities offer potential long-term cost and time efficiency [4].                                                           | The modular testing system enables rapid prototyping and iterative design.                                                                                           |
| Use of sustainable aviation fuel (SAF) can reduce carbon dioxide emissions by as much as 80% compared to fossil-based jet <sup>(10)</sup> , which aligns with future regulatory and industry trends. | Increased accessibility to aerodynamic testing, lowering barriers for universities, researchers, and developers who lack access to high-end wind tunnel facilities.  |
| Advanced active control systems (e.g. AFS, GLA) expand the safe flight envelope and allow operation of flexible or unconventional configurations [5].                                                | An increased focus on aeroelasticity and lightweight structures through the use of HAR wings and next-generation active control systems [2].                         |
| High-aspect-ratio (HAR) wings reduce induced drag, improving fuel efficiency and increasing available test endurance [2].                                                                            | Recent growth in autonomous and unmanned systems supports wider adoption of UAV-based testing platforms [6].                                                         |

<sup>10</sup> [The benefits of sustainable aviation fuel explained](#) (accessed on 29.04.2026)

**Table 3.3:** *Weaknesses and Threats of the Aircraft*

| Weaknesses                                                                                                                                                                                                                             | Threats                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Differences in drone aviation standards across countries stem from factors such as terrain, infrastructure, and altitude, making global standardization of aviation rules and regulations a challenge [7].                             | Airspace restrictions, certification requirements, and safety regulations may limit operation [7].                                        |
| The modularity of the test bed increases overall system complexity.                                                                                                                                                                    | Combining modular payloads, advanced controls, and measurement systems may lead to unforeseen system-level issues.                        |
| Each new test article introduces uncertainty in mass distribution, aerodynamics, and control response.                                                                                                                                 | Operational constraints such as weather dependency, limited test windows, and airspace access disrupt the testing schedule and frequency. |
| The use of sustainable aviation fuel (SAF) is difficult because it is currently expensive, limited in supply, and requires significant feedstock availability and infrastructure investment to compete with conventional jet fuel [8]. | Risk of vehicle loss during testing could increase cost, reduce project reliability, and pose a safety threat.                            |
| Due to environmental factors such as temperature fluctuations, humidity, and electromagnetic interference, in-flight measurements are noisier and face calibration challenges <sup>(11)</sup> .                                        |                                                                                                                                           |

After having performed the SWOT analysis, different categories can be combined: use strengths to exploit opportunities (SO), use opportunities to overcome weaknesses (WO), use strengths to mitigate threats (ST), consider both weaknesses and threats to minimise risk (WT).

<sup>11</sup> [Accuracy Isn't Enough: Why Measurement Uncertainty Matters in Aerospace Calibration](#) (accessed 29.04.2026)

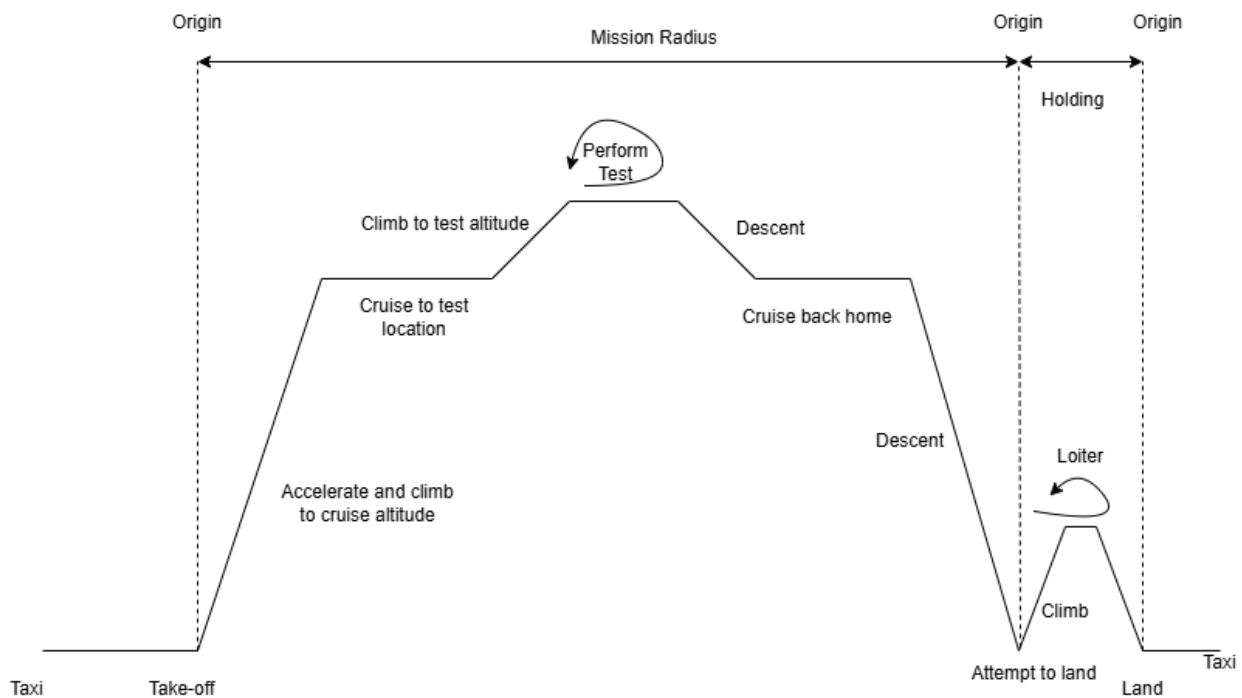
## 4. Functional Analysis

The first step to determine the requirements for the aircraft is looking at its function in detail. This can be done using a functional flow diagram (FFD) (see Figure 4.2), which analyses a typical mission of the aircraft. From that, key functions can be identified and later formatted as a functional breakdown structure (FBS) (see Figure 4.3).

The functions follow typical mission of the aircraft, and additional scenarios such as missed approach. The mission is subdivided in the Launch, Flight Test and Recovery phases. The first covers everything from pre-flight preparations up to and including the take-off, the second - the main phase of flight when the high-speed tests are performed, the third - descend, approach attempts and landing. Non-main success scenarios involved are missed approach and flight termination by command or when contact with the aircraft is lost. The flight termination method is to be determined when a particular design concept is selected.

### 4.1. Mission Profile

As the purpose of the aircraft is performing in flight experiments, it was decided that the mission profile will start and end at the same airport. Thus, the mission consists of the general phases that are stated in [9]. A diagram of the mission profile can be found in Figure 4.1.



**Figure 4.1:** General Mission Profile

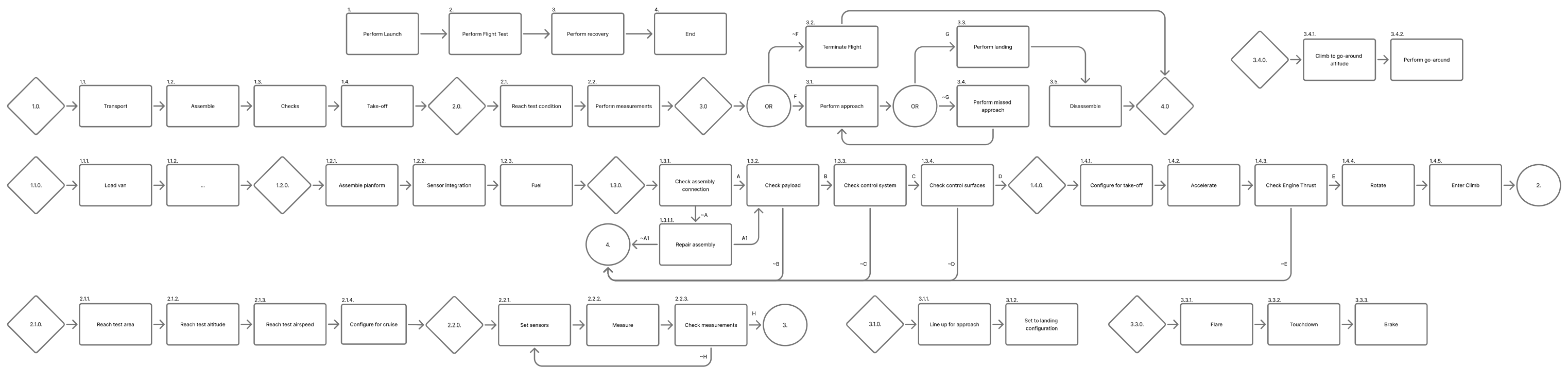


Fig. 4.2: Functional Flow Diagram

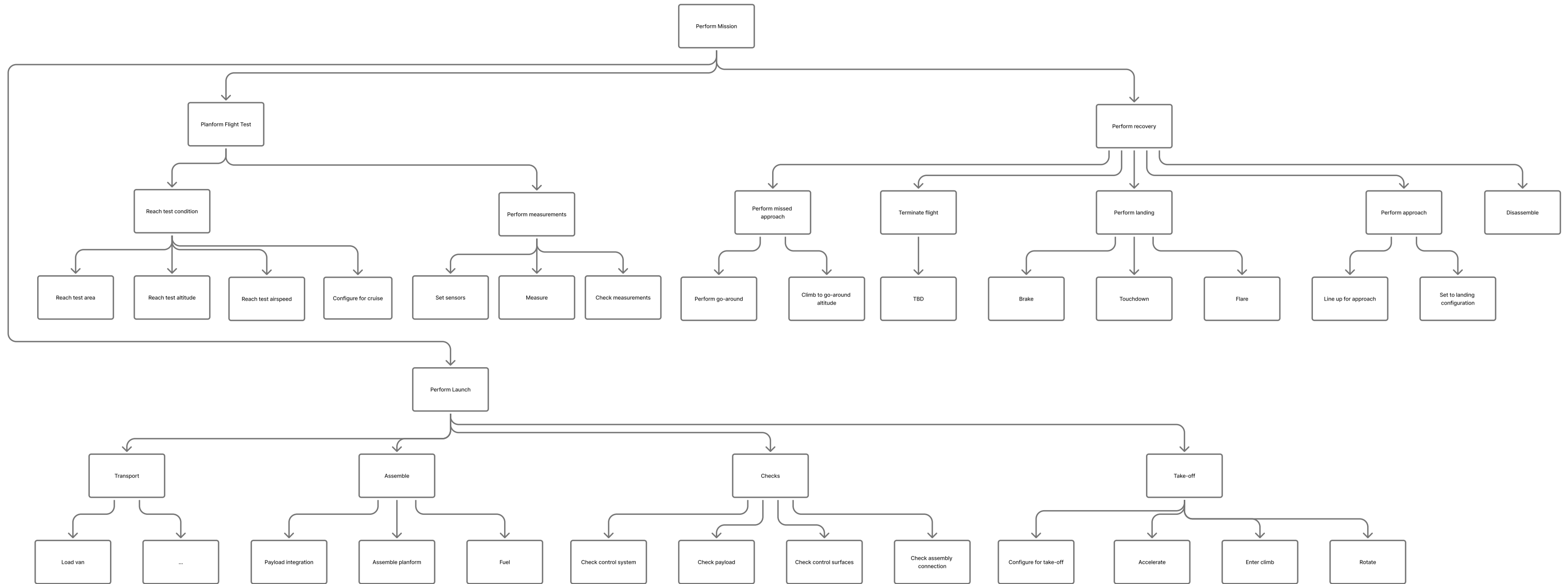


Fig. 4.3 Functional Breakdown Structure

## 5. Requirements and Constraints

Having evaluated the system functions, the requirements of the aircraft can be fully specified. To that end, a requirement discovery tree is created in [section 5.1](#).

### 5.1. Requirement Discovery Tree

In order to identify the system requirements a requirement discovery tree has been generated ([Figure 5.1](#)). The tree starts with the requirement for designing a flying model for testing flexible wings, which is split into the requirement to perform the flight test and the requirement to satisfy the constraints. The functional requirements stem directly from required functions of the design, as stated in the Functional Breakdown.

The constraints are split into (mass, payload volume, power, internal volume, communication link budget, van volume, cost, payload aerodynamic, c.g.), airworthiness requirements, manufacturability, lifetime, sustainability, autonomous flight capabilities and withstanding the aerodynamic loads.

### 5.2. Final List of System Requirements

[Table 5.1](#) lists the final system requirements divided by category: general (GEN), power (POW), communications (COM), control (CTL), propulsion (PRP), operations (OPS), certification (CER), financial (FIN) aerodynamics (AER), operations (OPS), certifications (CER), control (CON), stability (STA), and structures (STR). The driving requirements are written in red, and key requirements in italic.

**Table 5.1:** *List of System Requirements*

| Requirement ID     | Requirement                                                                                                                                                                              |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HUG-GEN-001        | The mass of the aircraft shall be less than 50kg, including payload [2].                                                                                                                 |
| HUG-GEN-002        | The available payload mass shall be equal to or larger than 5 kg [2].                                                                                                                    |
| HUG-POW-001        | The aircraft shall provide electrical energy necessary to power the payload, actuators, control subsystem, fuel subsystem and communication subsystem for the mission duration.          |
| HUG-POW-002        | The peak power provided by the aircraft shall be sufficient to provide full power to aircraft actuators, control subsystem, fuel subsystem and communication subsystem at the same time. |
| HUG-COM-001        | The aircraft shall transmit the measured data as well as telemetry from control, propulsion and fuel subsystems in real time from the maximum aircraft-ground station distance.          |
| HUG-COM-002        | The communication system shall allow for contact with the ground station for a distance equal to 1.5 up to the longer dimension of the testing area.                                     |
| HUG-CTL-001        | The aircraft shall have autonomous stabilisation systems.                                                                                                                                |
| HUG-CTL-002        | The aircraft shall be able to perform turns, climb and descent autonomously.                                                                                                             |
| HUG-CTL-003        | The aircraft shall be able to conduct experiments autonomously.                                                                                                                          |
| HUG-CTL-004        | The aircraft shall be remotely controllable beyond visual line of sight.                                                                                                                 |
| <i>HUG-CTL-005</i> | The c.g. excursion of the aircraft shall allow for both statically and dynamically stable, unstable and neutrally stable flight conditions.                                              |
| <b>HUG-CTL-006</b> | Isolated single-point failures of control surfaces or loss of thrust shall not cause catastrophic failures.                                                                              |
| <i>HUG-OPS-001</i> | All equipment for performing the flight test shall fit within a van.                                                                                                                     |
| HUG-OPS-002        | The aircraft shall be fully manufacturable using only TRL10 (Technology Readiness Level 10) manufacturing techniques.                                                                    |
| <b>HUG-OPS-003</b> | The aircraft shall be recyclable to at least an 80% degree by weight [10].                                                                                                               |
| HUG-OPS-004        | The endurance of the aircraft at maximum payload shall be equal to or longer than 30 minutes [2].                                                                                        |

| Requirement ID     | Requirement                                                                                                                                                                 |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>HUG-OPS-005</b> | The aircraft shall be able to sustain Mach 0.75 continuously for 5 min. in level flight                                                                                     |
| HUG-OPS-006        | The aircraft shall be able to sustain flight at Mach 0.4 during the entire duration of the main flight stage.                                                               |
| HUG-OPS-007        | The aircraft shall have sufficient fuel reserves to perform four landing attempts on top of the fuel required for maximum endurance.                                        |
| HUG-OPS-008        | The UAV shall use SAF as fuel source.                                                                                                                                       |
| HUG-OPS-009        | The UAV system shall be capable of conducting four test flights a day, in at least two different configurations.                                                            |
| HUG-CER-001        | The aircraft shall comply with airworthiness SORA requirements.                                                                                                             |
| HUG-CER-002        | It shall be possible to terminate the mission in a controlled manner at any point in time, using a ballistic parachute.                                                     |
| HUG-CER-003        | In case of communication loss, the aircraft shall autonomously return to the launch site.                                                                                   |
| <b>HUG-FIN-001</b> | The cost of a single aircraft configuration shall be less than 45000€ [2].                                                                                                  |
| <b>HUG-AER-001</b> | The critical Mach number of all aerodynamic surfaces shall be higher than Mach 0.75.                                                                                        |
| <b>HUG-AER-002</b> | The aircraft shall house test platforms of effective aspect ratios in the range between 19 to 27 [11].                                                                      |
| HUG-AER-003        | The aircraft shall house test platforms with taper ratios between 0.1 to 0.5.                                                                                               |
| HUG-AER-004        | The overall test platform shall see clear airflow during flight.                                                                                                            |
| HUG-AER-005        | The aircraft shall be able to perform the same manoeuvres (climb, turns, pull-up/down, dive) as required for CS-25 certification.                                           |
| HUG-STR-002        | The aircraft structure shall provide the internal volume for the fuel, propulsion, electrical, communications, and command subsystems.                                      |
| <b>HUG-STR-003</b> | The aircraft structure shall withstand stresses due to high subsonic loads at Mach 0.75, without yielding and with a safety factor of 1.5 for fracture.                     |
| HUG-STR-004        | The aircraft structure shall not buckle due to the aerodynamic loads at Mach 0.75.                                                                                          |
| <i>HUG-STR-005</i> | There shall be no wing-independent flutter or aeroelastic divergence of the baseline structure up to critical Mach number at the lowest operating altitude of the aircraft. |
| <b>HUG-STR-006</b> | The VULCAN engine(s) shall be integrated within the aircraft structure.                                                                                                     |
| HUG-STR-007        | The VULCAN engine shall be replaceable.                                                                                                                                     |
| HUG-STR-008        | The structure shall provide space and mounting for auxiliary systems for the VULCAN engine.                                                                                 |
| HUG-STR-009        | The structure shall include fuel tanks for housing the fuel-volume required for the entire mission, including missed approach and loiter time.                              |
| HUG-STR-010        | Every subsystem inside the aircraft wing and fuselage shall have the possibility of access, allowing for subsystem inspection.                                              |
| HUG-STR-011        | The inspection shall allow for identifying cracks in all primary load-bearing components.                                                                                   |
| <b>HUG-STR-012</b> | The aircraft shall retain structural integrity after 15 years of usage with four flights per day, 100 days a year                                                           |
| HUG-STR-013        | The natural frequencies of the structure shall not be within 0.8 and 1.4 [12] of the blade-pass frequency of the engine [13].                                               |
| HUG-STR-014        | The structure shall accommodate space for a secondary flight computer.                                                                                                      |
| HUG-STR-015        | The structure shall accommodate a payload bay of at least 5 litres.                                                                                                         |
| <b>Legend</b>      | <i>Key requirement</i> , <b>Driving requirement</b>                                                                                                                         |

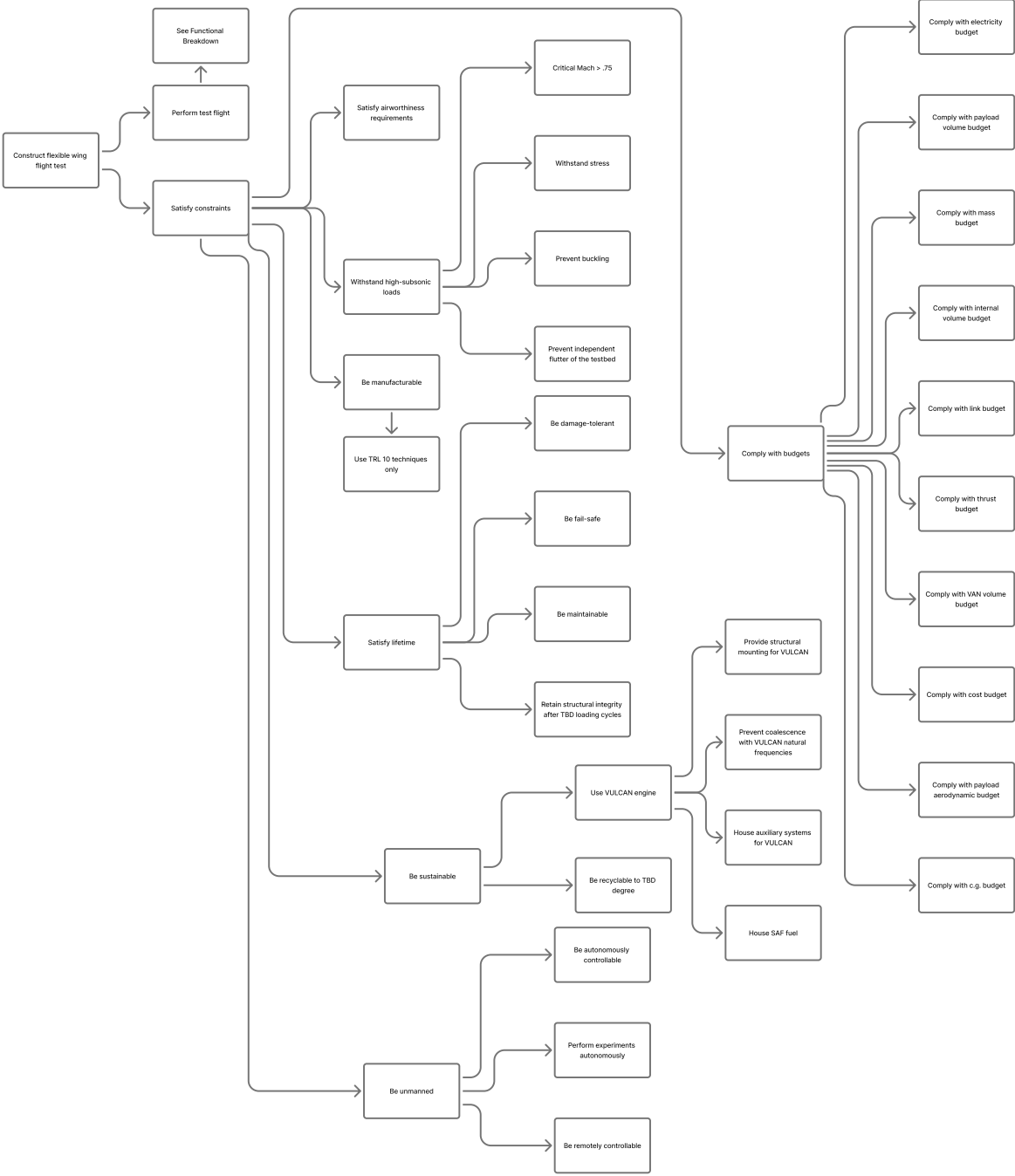


Figure 5.1: Requirement Discovery Tree

## 6. Resource Allocation

Resource allocation transforms the project objectives into a practical plan for execution by assigning the available time, personnel, budget and technical means to the activities required for the baseline phase. Its purpose is to ensure that the effort of the team is distributed in a structured and realistic manner, such that the major deliverables can be completed within the constraints of the project requirements. This chapter identifies the key resources available to the team, explains how these resources are divided across the main work packages, and highlights the rationale behind this distribution in view of project priorities, dependencies and expected workload.

### 6.1. Identifying The Available Budget

In this section, the general resources are identified and categorised, as such, a structure of the available budget can be made so that tracking each resource is possible. The budget originates from the restrictions set by the requirements, technical limitations and human availability. The structure of the resource allocation is based on the flow created by IBM <sup>(12)</sup>. The critical areas with a finite budget are:

- Centre of gravity location (CG location)
- Communication link budget
- Mass
- Aircraft dimensions
- Internal dimensions
- Fuel budget
- Power from the battery

#### 6.1.1. Centre of Gravity (CG) Location

The CG range of an aircraft is crucial for its stability and controllability limits. An unbalanced vehicle cannot guarantee safety or functionality during operations. The specific margins of stability and controllability will be subject to change based on the payload and desired test conditions. The specific CG range can be evaluated based on the following:

- Longitudinal position of the wing w. r. t. the fuselage,
- Tail size and position,
- Distribution and location of on-board systems,
- Distribution and location of payload,
- Location of fuel tanks and batteries,
- Location and configuration of landing gears,
- Implementation of a control system for the fuel flow.

Verifying the CG position against the allowable limits can be done with the use of a scissor plot combining the stability and controllability curves, which in turn defines the required tail size and longitudinal wing position. The process can be iterated until a balanced design is achieved that can operate within the desired stability and controllability limits.

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<sup>12</sup>[IBM Resource Allocation](#)(accessed on 24.04.2026)

### 6.1.2. Communication Link

An analysis of the link budget is required to ensure communication with the UAV can be performed successfully during all operations. The main design considerations stem from mission range and endurance, as well as operating conditions of the UAV. In particular, the following parameters are used to define the communication link budget:

- Power available (supplied by batteries),
- Transmitter/Receiver power,
- Maximum UAV range during operations,
- Maximum UAV velocity,
- Available volume and internal component arrangement.

The nominal mission will last 30 minutes, with the test phase accounting for 5 minutes of the total time. The speed at the test-phase will be 0.75 Mach. At this speed, keeping a clear and low-latency communication link will be the main requirement that will drive the design of the communication system.

### 6.1.3. Mass

Mass is a restricted resource that must be monitored throughout the development process. This constrain is derived from the stakeholder requirement (**HUG-GEN-001**), limiting the mass of the drone to 50 kg. This maximum design mass includes the imposed minimum internal load capacity of 5 kg (**HUG-GEN-003**), onboard fuel, and all aircraft subsystems. The mass is further constrained by the propulsion system. From the requirements, it is known that one engine weighs 2.1 kg, and if the design requires 2 such power plants, a total of 4.2 kg is allocated to this system. Accounting for the mass of cables and hoses needed, it would result in a combined mass of 5 kg.

The rest of the mass is allocated towards the rest of the components that completes the drone. The remaining 40kg are allocated towards:

- Fuel
- Wings assembly
- Fuselage assembly
- Tail assembly
- Internal battery assembly

Although the power plant is installed within the fuselage, its mass is treated as a separate contribution from the fuselage assembly. The fuselage mass budget therefore, includes the structural shell, internal supports, and fuel tanks, but excludes both the propulsion-system mass and the fuel mass itself. Cables and connectors used to transfer power from the battery to the payload are considered in the battery assembly.

The general idea of modularity will permit customers to modify the allocation of mass to comply with the user test case. Mass is coupled with the location of the centre of mass, as such, both resources have to be closely monitored together.

### 6.1.4. Aircraft dimensions

There is a set restriction for the outer dimensions of the aircraft set by the requirement that it and all necessary equipment to operate it should fit inside a van. This is something that should be taken into consideration while designing the plane.

To ensure a more sustainable transport of the aircraft the electric Renault master E-tech L3H2 variant was chosen as reference van <sup>(13)</sup>.

**Table 6.1:** Renault Master E-Tech L3H2 Loading Specifications

| Specification                        | Value             |
|--------------------------------------|-------------------|
| <b>Loading Area — Panel Van L3H2</b> |                   |
| Usable length                        | 3733 mm           |
| Usable length at 1.1 m               | 3680 mm           |
| Usable width                         | 1765 mm           |
| Usable height                        | 1894 mm           |
| <b>Sliding Side Door(s)</b>          |                   |
| Width of sliding side door(s)        | 1270 mm           |
| Height of sliding side door(s)       | 1780 mm           |
| <b>Rear Door</b>                     |                   |
| Width (169 mm from floor)            | 1580 mm           |
| Entrance height                      | 1820 mm           |
| Cargo threshold (min/max)            | 543/557 mm        |
| Loading volume                       | 13 m <sup>3</sup> |

Optimizing the packing of the disassembled aircraft might become necessary on later stages in the design process.

### Internal dimensions

For the aircraft to function as intended several components need to be present within the aircraft:

- Fuel tanks and fuel system
- Engine system
- Communication system
- Onboard computer
- Payload
- Power system

Not only does the aircraft need to accommodate all of these systems but some of the components will be highly location dependent (for example the engine system). This is all something that must be taken into account when designing the aircraft.

#### 6.1.5. Fuel Budget

The amount of fuel onboard the plane has to be enough to sustain the 30-minute flight requirement (25 min cruise and 5 min testing), along with facilitating the take-off and landing procedures. In addition to the mandatory mission segments, the drone requires a reserve of fuel to allow loitering for a period of 5 minutes, as stated by **HUG-OPS-004**.

As stated in [Table 5.1](#), requirement **HUG-SUS-001**, the drone will use SAF (sustainable aviation fuel) as a fuel source. During the design process of the propulsion and fuel systems, a choice on the type of fuel has to be made. The primary factors that influence the fuel consumption, and as such the fuel required are:

<sup>13</sup> [Renault Master Van](#) (accessed on 24.04.2026)

- Mass of the drone
- Consumption of the engine
- Altitude of operation

#### 6.1.6. Power available

An analysis of the internal payload and electronic subsystem revealed the necessity of a battery. This raises the need for a power budget, as a bigger battery will take from the mass budget of other subsystems. The battery will be part of the base drone and will not count towards the payload mass. It will have a specific power available that the payload can access. The main parts that affect the power of the battery are:

- Communication system
- Power plant system
- Payload
- Internal electric system of the drone

#### 6.1.7. Human factor

For the completion of the design phase, a team of 10 people is employed. They are further separated into 4 departments that will work alongside each other, promoting concurrent engineering. This method is used by Airbus [14], and has proven to have the highest production rate and outcomes. Each one of the departments will make use of the resources available, as explained in this section. The design team are divided into the following departments: Structures, Aerodynamics, Propulsion and Control & Systems.

### 6.2. Strategic Resource Planing

Strategic planning involves identifying resource needs across departments, teams and project plans. It requires the clarification of long and short-term project goals and schedule resource allocation according to strategic objectives to prevent resource underusage and overallocation.

This implies usage tracking along with development, collaboration and communication between departments and dynamic reallocation in case of misuse of resources.

The distribution of resources depends on the needs of each department and their specialisation, teams having to share some of them. The table contains the allocation of each team and their associated needed resource:

**Table 6.2:** *Departmental responsibility matrix.*

| Department             | CG loca-<br>tion | Link bud-<br>get | Mass | Aircraft dimen-<br>sions | Internal di-<br>mensions | Fuel bud-<br>get | Power from bat-<br>tery |
|------------------------|------------------|------------------|------|--------------------------|--------------------------|------------------|-------------------------|
| Structures             | ×                |                  | ×    | ×                        | ×                        |                  |                         |
| Aerodynamics           | ×                |                  | ×    |                          |                          |                  |                         |
| Propulsion             |                  |                  | ×    |                          | ×                        | ×                | ×                       |
| Control & Sys-<br>tems | ×                | ×                |      | ×                        | ×                        |                  | ×                       |

This initial distribution is susceptible to change as the development process advances and unforeseen circumstances appear.

### 6.2.1. Usage Tracking

Effective resource allocation requires continuous monitoring of how each resource is used throughout the design process. This is necessary to ensure that no department exceeds its allocated budget and that changes in one subsystem are communicated to all affected departments. Since resources such as mass, internal volume, fuel, battery power and CG location are strongly coupled, they cannot be tracked independently by each department without a common overview.

Resource tracking will therefore be performed using a centralised budget file, accessible to all departments. This file will contain the latest estimated value, allocated limit, remaining margin and responsible department. This ensures that all departments work with the same baseline values and prevents outdated assumptions from propagating through the design.

The mass budget, fuel budget and battery power budget will be updated whenever a subsystem mass, fuel consumption estimate or electrical power requirement changes. The CG location will be recalculated after relevant changes in component placement, fuel tank location, payload position or battery arrangement. Similarly, internal and external dimensions will be updated when changes are made to the fuselage layout, modular wing interface, payload bay or transport configuration.

To support concurrent engineering, resource usage will be discussed during regular team meetings. If a resource exceeds its allocated limit, or if the margin becomes too small, the responsible department must notify the affected departments and propose a corrective action. Possible actions include redesigning the subsystem, reallocating budget from another department, changing the component placement, or revising the configuration layout. Major changes that affect multiple departments shall be documented as design decisions to maintain traceability.

This approach ensures that resource usage remains visible throughout the project and that design changes are evaluated at system level rather than only within individual departments. As a result, the team can identify budget conflicts early and prevent late-stage integration issues.

## 7. Preliminary Configuration Development

This chapter details a systematic, top-down approach employed to develop design concepts for a high-speed aeroelastic testing platform. The approach employed utilizes Design Option Trees (DOT) to present possible solutions, shown in [Figure 7.1](#). First, top-level architectural designs are presented in [section 7.1](#), to later determine the vehicle category best suited to achieve the project objectives whilst adhering to the requirements specified in [Table 5.1](#). Following the selection of a vehicle category, a second DOT is used to explore subsystem options. Finally, preliminary configuration concepts are generated in [section 7.3](#), yielding a shortlist of viable strawman design concepts to be further analysed in the detailed trade-off phase.

### 7.1. Top-Level Platform Selection

Before evaluating specific aircraft subsystems, an initial design option tree was generated to determine the overarching vehicle architecture. This top-level tree functions as an "OR Tree" where the primary path is selected from distinct, mutually exclusive operational concepts. Three main approaches were evaluated to perform aeroelastic experiments mid-flight: a replaceable planform on a dedicated drone, a per-test custom-made drone, and designing an invariant aircraft capable of mounting test specimens.

- **Replaceable planform:** Accepted for further development due to high versatility. In relation to the alternative platforms, one with a replaceable planform was selected for further conceptualisation as it was immediately evident that it would allow for the widest variation of experiments.
- **Test-specific custom design:** Rejected due to existence of preferable alternatives. A test-specific custom design was considered as it would represent less structural complexity in relation to a fully replaceable planform. The concept was not further developed as it was deemed that generating a new design and manufacturing iteration, and possibly re-certifying the aircraft for each experiment, was significantly more complex and wasteful than overcoming structural complexities.
- **Mounting on invariant aircraft:** Rejected due to existence of preferable alternatives. An invariant aircraft with mounting ports was suggested as an option to carry out aeroelastic experiments. Benefits considered included a relative simplicity stemming from the invariant planform, but the concept was ultimately not developed further as it was deemed that it was too restrictive to the range of experiments that could be carried out upon it, in relation to the varying planform alternative.

For each of these top-level approaches, various vehicle types were considered, including fixed-wing aircraft, copters (both helicopters and multicopters), Vertical Take-Off and Landing (VTOL) fixed-wing aircraft, and towing aircraft.

- **Rotary-Wing Platforms (Helicopters & Multicopters):** Rejected across all branches due to requirement violations. Rotary wing platforms were proposed due to "tried and true" excellence in modularity and payload flexibility, yet their style of flight makes high-speed subsonic flight unfeasible, directly in violation of requirement (HUG-OPS-005) [[15](#)].
- **VTOL Fixed-Wing:** Rejected across all branches due to mass and complexity penalties. This concept was proposed to increase operational flexibility by increasing the number of suitable take-off and landing locations. However, in addition to increasing design complexity, the addition of a tilting mechanism for the engines would rapidly consume the tight 50kg mass budget (HUG-GEN-001). [[16](#)].
- **Towing Aircraft:** Rejected as an invariant platform due to requirement violations. This concept aimed to reduce the mass of the flying laboratory by removing its independent thrust components, and instead obtaining thrust from a separate plane which tows it. The concept was initially rejected due to the expected disturbance of the tow plane's wake to the airflow seen by the test article (HUG-AER-004) [[9](#)]. Furthermore, it was deemed that this concept would risk the tow plane during flutter testing if it became uncontrollable which would violate Specific Operations Risk Assessment (SORA) range safety limits (HUG-CER-001, HUG-CER-002) [[1](#)].

- **Conventional Fixed-Wing:** Accepted across all branches. This concept was deemed as the only viable architecture capable of achieving Mach 0.75, whilst maintaining clean airflow for test specimens (HUG-OPS-005, HUG-AER-001). Furthermore, a conventional fixed-wing drone was most conducive to predictable, decoupled flight dynamics necessary for baseline flutter suppression testing (HUG-CTL-001, HUG-STR-005) [5, 9, 16].

From this analysis, the drone architecture was narrowed down to a conventional fixed-wing design.

## 7.2. Subsystem Design Option Tree

Having established a conventional fixed-wing approach, a second detailed DOT was generated. Unlike the top-level selection, this fixed-wing tree acts as an "AND Tree" where valid configurations are built by combining accepted options from each fundamental subsystem group (Fuselage, Empennage, Wing, Propulsion, and Undercarriage).

- **Rejected Concepts (Red):** Options eliminated by unanimous group consensus. Rejections were strictly based on obvious technical infeasibility or direct conflicts with the top-level project requirements.
- **Concepts Considered Further (CCF) (Orange):** Options which could potentially work, yet appear to be unnecessarily complex in relation to alternatives, or it is not possible to analyse their feasibility within the time-frame of this project. These options were considered but not developed further into strawman concepts.
- **Accepted Concepts (Green):** Options deemed feasible and directly employed towards strawman concept development.

**Table 7.1:** *Sub-System Options and Initial Screening*

| Subgroup                       | Option    | Status   | Justification                                                                                                                                                                                                                                                                                       |
|--------------------------------|-----------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Fuselage Group</b>          |           |          |                                                                                                                                                                                                                                                                                                     |
| Internal Construction Material | Composite | Accepted | Offers the high strength-to-weight ratio necessary to survive the high subsonic loads at Mach 0.75 (HUG-STR-003, HUG-OPS-005) while staying under the strict 50kg mass requirement (HUG-GEN-001) [16].                                                                                              |
|                                | Metal     | Accepted | Traditional aerospace manufacturing method; provides predictable structural dynamics and ease of modular manufacturing using TRL10 techniques (HUG-OPS-002), though potentially heavier.                                                                                                            |
|                                | Polymer   | Rejected | Will generally lack the required structural stiffness and yield strength to act as the primary load-bearing core for a high subsonic aircraft (HUG-STR-003). Higher performing polymers are usually much harder to produce. [16].                                                                   |
| Skin Material                  | Composite | Accepted | Can provide an aerodynamically smooth surface finish necessary for high speed subsonic flight and high skin stiffness to prevent buckling (HUG-STR-004). Furthermore, composites are typically characterised with a high specific modulus, which is favourable in order to minimise mass.           |
|                                | Metal     | Accepted | Highly durable and relatively simple to integrate (HUG-OPS-002), though harder to form into complex aerodynamic curves required for clean airflow (HUG-AER-004) [9, 16]. Tends to have lower specific modulus compared to fibre reinforced composite materials, which would incur a weight penalty. |

*Continued on next page*

| Subgroup                | Option             | Status   | Justification                                                                                                                                                                                                                                                                                                                                          |
|-------------------------|--------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | Polymer            | CCF      | Feasible for non-load-bearing fairings and lightweight sections, but highly susceptible to flutter and buckling under high dynamic pressures (HUG-STR-004, HUG-STR-005) [16].                                                                                                                                                                          |
| Avionics                | Manual & Autopilot | Accepted | Satisfies the requirement for a remotely controlled (HUG-CTL-004), autonomous testbed (HUG-CTL-001, HUG-CTL-003), allowing a pilot to intervene and terminate the mission in a controlled manner during critical test points or emergencies (HUG-CER-003).                                                                                             |
|                         | Autopilot Only     | CCF      | Feasible for pure drone operations, but presents safety and recovery risks during the induced flutter testing required by the project, potentially hindering controlled mission termination (HUG-CER-003).                                                                                                                                             |
|                         | Manual Only        | Rejected | Directly violates the core project requirement of designing an autonomous flying laboratory (HUG-CTL-001, HUG-CTL-003), and human reaction times are insufficient to manually suppress high-frequency flutter [1].                                                                                                                                     |
| <b>Electrical Group</b> |                    |          |                                                                                                                                                                                                                                                                                                                                                        |
| Electrical Subgroup     | Battery            | Accepted | A proven, reliable, and independent energy source that can quickly be sized to meet voltage and capacity requirements. Current available battery technology has relatively poor energy density, thus alternative power options might prove to be lighter.                                                                                              |
|                         | Power Generator    | Accepted | Commercial jet turbines exist that are able to use their starter motor as an alternator in flight and provide up to 900 Watts of continuous power, which could support all the subsystems of the aircraft (HUG-OPS-004). <sup>(14)</sup>                                                                                                               |
| <b>Empennage Group</b>  |                    |          |                                                                                                                                                                                                                                                                                                                                                        |
| Canard Modularity       | Permanent          | CCF      | While structurally feasible, a permanent canard is an unconventional baseline that severely restricts the platform's versatility, going against the main mission objective. It permanently disrupts upstream airflow (HUG-AER-004), making it difficult to collect clean baseline data for standard, non-canard commercial architectures [16, 17].     |
|                         | Static Port        | Accepted | A method to test specific canard configurations only when necessary, or to test smaller wing profiles in clean air (HUG-STR-003, HUG-AER-004) [17]. <b>we actually can't do that (1:1 scale)!</b>                                                                                                                                                      |
|                         | Variable Port      | CCF      | The canard is not the primary experimental article for this project. Engineering a variable-position mounting mechanism would introduce significant complexities and possible structural weaknesses, along with mass penalties (HUG-STR-003, HUG-GEN-001) that far outweigh any secondary aerodynamic benefits, detracting from the core mission [16]. |

*Continued on next page*

<sup>14</sup>JetCat Turbine with Generator (accessed on 30.04.2026)

| Subgroup          | Option                                       | Status   | Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------|----------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | No port                                      | Accepted | Designing an aircraft which cannot test canard configurations limits the utility of the UAV, however, simplifies design, allows for a lighter structure and increases the overall payload volume, which may be crucial to satisfy the payload requirement, (HUG-STR-015) as well as having low enough drag to satisfy the velocity requirement, (HUG-OPS-005)                                                                                                                                                                                                                |
| Tail Mod-ularity  | Static Port                                  | Accepted | Allows swapping of empennage geometries to allow for more varied testing, without excessive structural weight or compromising the primary load paths (HUG-STR-003) [9]. Having a port would still allow the attachment of a basic tail-cone to simulate a flying wing configuration, if desired.                                                                                                                                                                                                                                                                             |
|                   | No port                                      | Rejected | Required only for flying wing configurations (HUG-STR-017) or specific tailless baseline evaluations, however makes it impossible to test configurations which require a tail.                                                                                                                                                                                                                                                                                                                                                                                               |
|                   | Permanent                                    | Accepted | Conventional, highly rigid, lightweight approach, ensuring clean data collection from the main wing (HUG-STR-005), but limits versatility of the testing. [1].                                                                                                                                                                                                                                                                                                                                                                                                               |
| Tail Con-figs.    | T, H, V, In-verted V, Con-ventional, No tail | Accepted | All represent valid aerodynamic trade-offs between clean airflow (HUG-AER-004), drag limits (HUG-OPS-005), and structural complexity that require configuration based evaluation [9, 16].                                                                                                                                                                                                                                                                                                                                                                                    |
|                   | Twin Boom                                    | Rejected | Introduces excessive structural weight and its own complex aeroelastic modes, heavily penalizing versatility. Additionally this is an unconventional design, not relevant to the future of general aviation design. [16].                                                                                                                                                                                                                                                                                                                                                    |
| <b>Wing Group</b> |                                              |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Wing Con-figs.    | Low, Mid, High                               | Accepted | Standard configurations offering different balances of payload capacity and volume (HUG-GEN-002, HUG-STR-015, HUG-STR-002), and other configuration level trade offs. [16].                                                                                                                                                                                                                                                                                                                                                                                                  |
|                   | Flying Wing                                  | CCF      | While this configuration presents high aerodynamic efficiency (HUG-OPS-005) [16], and experimental aircraft like the X-56 have shown the viability of this configuration [2], its versatility as a testbed is limited (HUG-AER-002, HUG-AER-003). The lack of a distinct empennage prevents the testing of many coupled baseline flutter modes [1]. Furthermore, mounting an experimental planform on a flying wing requires placing it directly above or below the main body, which would demand highly control systems, straining (HUG-CTL-001, HUG-CTL-002, HUG-CTL-003). |
|                   | Delta                                        | Rejected | Inherently stiff and low-aspect-ratio [16], directly conflicting with the mission to test highly flexible HAR wings (HUG-AER-002) [18].                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                   | Prandtl Plane                                | Rejected | The bell-shaped lift distribution introduces aeroelastic behaviors that are too esoteric for a baseline modular testbed aiming to validate standard control systems (HUG-CTL-001) [19].                                                                                                                                                                                                                                                                                                                                                                                      |

*Continued on next page*

| Subgroup                | Option                  | Status   | Justification                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------|-------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Modularity              | Static Port             | Accepted | Lightweight and simple method for swapping test articles (HUG-AER-003) at a fixed aerodynamic center [16]. The fixed location of the conceptualised ports would not limit experimental planform characteristics such as root chord or root thickness.                                                                                                                           |
|                         | Variable Port (X)       | Accepted | Allowing longitudinal shifting of the wings provides flexibility for aerodynamic center management (HUG-CTL-005) when testing different planforms [9].                                                                                                                                                                                                                          |
|                         | Variable Port (Z)       | CCF      | Being able to change the location of the primary load-bearing wing vertically would likely introduce severe fuselage reinforcement mass penalties to avoid structural weaknesses (HUG-GEN-001, HUG-STR-004) [16].                                                                                                                                                               |
|                         | Variable Port (X & Z)   | CCF      | The combined complexity guarantees exceeding the 50kg limit (HUG-GEN-001) and drastically reduces primary structural fracture safety margins (HUG-STR-003) [16].                                                                                                                                                                                                                |
|                         | Replaceable Mid Section | CCF      | Segmenting the fuselage core weakens the primary structure (HUG-STR-003) and complicates avionics routing (HUG-STR-002) [14].                                                                                                                                                                                                                                                   |
|                         | Permanent               | Rejected | Violates the central requirement to house and swap varying test platforms (HUG-AER-003).                                                                                                                                                                                                                                                                                        |
| <b>Propulsion Group</b> |                         |          |                                                                                                                                                                                                                                                                                                                                                                                 |
| Mounting Location       | Wing-mounted            | CCF      | Introducing concentrated masses to highly flexible test wings would heavily alter specific flutter modes (HUG-STR-005), and it would also complicate the modularity and exchangeability of the wings (HUG-AER-003) [1]. This concept was originally considered as it would be a representation of what future commercial transportation aircraft are assumed to look like [11]. |
|                         | Buried engine           | CCF      | Aerodynamically efficient, minimizing drag for high subsonic speeds (HUG-OPS-005), however the surface temperatures of jet engines requires having large air gaps around the engine, which complicates the design. [16, 20].                                                                                                                                                    |
|                         | Fuselage mounted        | Accepted | Centralises mass for stability (HUG-CTL-005) and leaves wings clear for unobstructed HAR testing (HUG-AER-004) [9]. Would likely limit the variety of vertical tails that can be tested.                                                                                                                                                                                        |
| Fuel Compatibility.     | SAF-only                | Accepted | Complies with the sustainability requirement mandating Sustainable Aviation Fuel usage (HUG-OPS-008) [21].                                                                                                                                                                                                                                                                      |
|                         | SAF Blend               | Accepted | Provides operational flexibility during early testing phases while still supporting the transition to sustainability (HUG-OPS-008) [8, 22].                                                                                                                                                                                                                                     |
|                         | JET-A1                  | Rejected | Fails the core sustainability requirement mandating SAF usage (HUG-OPS-008) [21].                                                                                                                                                                                                                                                                                               |
| Modularity              | Fixed                   | Accepted | Standard, highly reliable integration (HUG-CTL-006) of the propulsion system, minimizing structural vibration coupling (HUG-STR-013) [12, 13].                                                                                                                                                                                                                                  |

*Continued on next page*

| Subgroup                   | Option            | Status   | Justification                                                                                                                                                                                                                                                          |
|----------------------------|-------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | Quick Swap-pable  | CCF      | The complexity of quick-swappable thrust-bearing mounts adds unacceptable risk and structural weight (HUG-GEN-001). This goes against the recommendation to implement intrinsic reliability in the design of UAVs. [23].                                               |
| Fuel Tank Loc.             | Wing and Fuselage | CCF      | Small scale HAR wings will likely not have sufficient internal volume to warrant the design complexity of fuel storage and plumbing in the wings of an aeroelastic, swappable wing.                                                                                    |
|                            | Fuselage Only     | Accepted | Keeps CG stable (HUG-CTL-005) and allows test wings to remain slender without internal fuel volume constraints, crucial for HAR designs (HUG-AER-002, HUG-STR-009) [18].                                                                                               |
|                            | Wing Only         | Rejected | The flexible, HAR test wings lack the internal volume necessary to store sufficient fuel for a 30-minute endurance (HUG-OPS-004, HUG-STR-009) [18].                                                                                                                    |
| <b>Undercarriage Group</b> |                   |          |                                                                                                                                                                                                                                                                        |
| Mounting points            | On wings          | CCF      | Mounting the landing gear on High AR aeroelastic wings is challenging due to the inherent low stiffness and the added internal volume requirements. Additionally, swapping the wings would then imply having to swap out the landing gear as well. (HUG-AER-002) [16]. |
|                            | On fuselage       | Accepted | Keeps the test wings entirely unencumbered by gear structures, allowing unrestricted bending profiles (HUG-STR-005) [9].                                                                                                                                               |
| Structure                  | Retractable       | Accepted | Essential for reducing parasitic drag to achieve Mach 0.75, though comes with some additional complexity and weight. (HUG-OPS-005) [16].                                                                                                                               |
|                            | Skids             | Accepted | Drastically reduces weight (HUG-GEN-001) and cost (HUG-FIN-001), though requires external launch infrastructure, or pogos that detach during takeoff, as seen on the U2. [24]                                                                                          |
|                            | Fixed             | Rejected | The drag penalty makes speed (HUG-OPS-005, HUG-AER-001) and endurance (HUG-OPS-004) targets virtually impossible to meet simultaneously [16].                                                                                                                          |

### 7.3. Generation of Preliminary Configuration Concepts

Now that the design option tree has been screened to only allow for plausible subsystem options, potential aircraft configurations must be generated for holistic evaluation. If every accepted option in the DOT were combined simultaneously, it would result in an impractical number of potential aircraft concepts. For this reason, the configuration selection process is split up into 2 stages to make the selection procedure more manageable. Subsystems such as avionics, materials, and specific tail geometries were omitted from the configuration generation at this stage of the project since they are agnostic to the overall aerodynamic and structural configuration of the aircraft.

Stage 1 focuses exclusively on defining the optimal level of modularity for the aircraft. Because one of the core objectives of the flying laboratory is to test highly flexible aircraft components, the degree of modularity drives the structural weight, internal volume constraints, and overall feasibility of the testbed. Once the ideal modularity baseline is established in Stage 1, Stage 2 will integrate these findings with the remaining aerodynamic, structural and propulsion subsystems to generate comprehensive aircraft configurations for the final trade-off.

### 7.3.1. Stage 1: Modularity Configuration Selection

The focus of this stage was strictly limited to the modularity branches of the design option tree: Wing Modularity, Canard Modularity, and Tail Modularity. Options that presented no distinct benefit, as well as CCFs, were previously pruned, leaving only the highly viable structural interfaces presented in [Table 7.2](#).

**Table 7.2:** *Stage 1: Modularity Design Options (Accepted Concepts)*

| Design Category              | Accepted Options               |
|------------------------------|--------------------------------|
| Wing: Modularity             | Static Port, Variable Port (X) |
| Empennage: Canard Modularity | Static Port, No port           |
| Empennage: Tail Modularity   | Static Port, Permanent         |

By systematically combining these accepted modularity parameters, eight distinct structural architectures were generated. These configurations range from the bare minimum modularity required to fulfil the mission objectives, to fully swappable architectures that maximize testing versatility. These options are detailed in [Table 7.3](#).

**Table 7.3:** *Stage 1: Preliminary Modularity Configurations*

| Config. ID  | Wing Modularity   | Canard Modularity | Tail Modularity | Verdict  |
|-------------|-------------------|-------------------|-----------------|----------|
| HUG-CFG-101 | Static Port       | No port           | Permanent       | Rejected |
| HUG-CFG-102 | Static Port       | No port           | Static Port     | Accepted |
| HUG-CFG-103 | Static Port       | Static Port       | Permanent       | Rejected |
| HUG-CFG-104 | Static Port       | Static Port       | Static Port     | Accepted |
| HUG-CFG-105 | Variable Port (X) | Static Port       | Permanent       | Rejected |
| HUG-CFG-106 | Variable Port (X) | Static Port       | Static Port     | Accepted |
| HUG-CFG-107 | Variable Port (X) | No Port           | Permanent       | Rejected |
| HUG-CFG-108 | Variable Port (X) | No Port           | Static Port     | Accepted |

From these modularity configurations, HUG-CFG-101, HUG-CFG-103, HUG-CFG-105, HUG-CFG-107 are not feasible due to a lack of tail adaptability. Having a permanent tail would significantly limit the flight envelopes, aircraft configurations, and stability conditions that could be analysed by researchers. HUG-CFG-102, HUG-CFG-104, HUG-CFG-106 and HUG-CFG-108 offer varying levels of configurability. Not having the ability to adjust the longitudinal position of the wing would require shifting the CG of the craft using a ballast to maintain stability, which would incur a weight penalty, but might be better than designing a structure capable of moving the wing location, so it must be considered further.

### 7.3.2. Stage 2: Aerodynamic Layout and Integration

Stage 2 of the configuration selection focusses on the rest of the options of the tree which drive the design. Since the fuselage and skin materials cannot be decided upon at this stage, and branches such as propulsion mounting, location and undercarriage mounting location have been narrowed down to one option each, they need not be considered here. Additionally the tail configuration no longer needs to be decided, since all the modularity configurations include the ability to swap the tail section to whichever configuration is most suitable for the mission.

As a result of this pre-screening, the driving parameters were successfully isolated to just two categories: Wing Vertical Position and Undercarriage type, as shown in [Table 7.4](#).

**Table 7.4:** *Stage 2: Aerodynamic and Geometric Design Drivers*

| Design Category         | Accepted Options              |
|-------------------------|-------------------------------|
| Wing: Vertical Position | Low Wing, Mid Wing, High Wing |
| Undercarriage Type      | Retractable Gear, Skids       |

These options were permuted to arrive at 6 options, as detailed in [Table 7.5](#). From these, most options can be pruned out for infeasibility or due to lack of merit. HUG-CFG-202 was rejected since low mounted wings are incompatible with skid landing, due to the inevitability of wing strikes during landing, especially considering the aircraft could be configured with high aspect ratio, flexible wings. Additionally, mid-wing configurations HUG-CFG-203 and HUG-CFG-204 were eliminated because a mid-mounted central wingbox would disrupt a continuous internal cavity for possible payload bays, fuel tanks, or internal components. Since fuselage mounted propulsion is the only remaining option in the tree, high wing concepts HUG-CFG-205 and HUG-CFG-206 would result in the engines ingesting the wake from the wings, diminishing their performance. This leaves us with only one option from stage 2, namely, HUG-CFG-201.

**Table 7.5:** *Stage 2: Initial Aerodynamic and Geometric Permutations*

| Aero Config. ID | Wing Position | Undercarriage    | Verdict  |
|-----------------|---------------|------------------|----------|
| HUG-CFG-201     | Low Wing      | Retractable Gear | Accepted |
| HUG-CFG-202     | Low Wing      | Skids            | Rejected |
| HUG-CFG-203     | Mid Wing      | Retractable Gear | Rejected |
| HUG-CFG-204     | Mid Wing      | Skids            | Rejected |
| HUG-CFG-205     | High Wing     | Retractable Gear | Rejected |
| HUG-CFG-206     | High Wing     | Skids            | Rejected |

Now, we can combine this configuration from stage 2, with all the configurations from stage 1 to arrive at the final set of configurations to analyse further, as shown in [Table 7.6](#). Note that all of the configurations feature a low wing, retractable landing gear, fuselage-mounted fuel tanks, and fuselage-mounted engines. The main difference lies in the modularity, which is dependent on the types of ports for the wing, canard, and tail.

**Table 7.6:** *Final Integrated Candidate Architectures for Detailed Trade-Off*

| ID          | Wing Modularity   | Canard Modularity | Tail Modularity |
|-------------|-------------------|-------------------|-----------------|
| HUG-CFG-301 | Static Port       | No Port           | Static Port     |
| HUG-CFG-302 | Static Port       | Static Port       | Static Port     |
| HUG-CFG-303 | Variable Port (X) | Static Port       | Static Port     |
| HUG-CFG-304 | Variable Port (X) | No Port           | Static Port     |

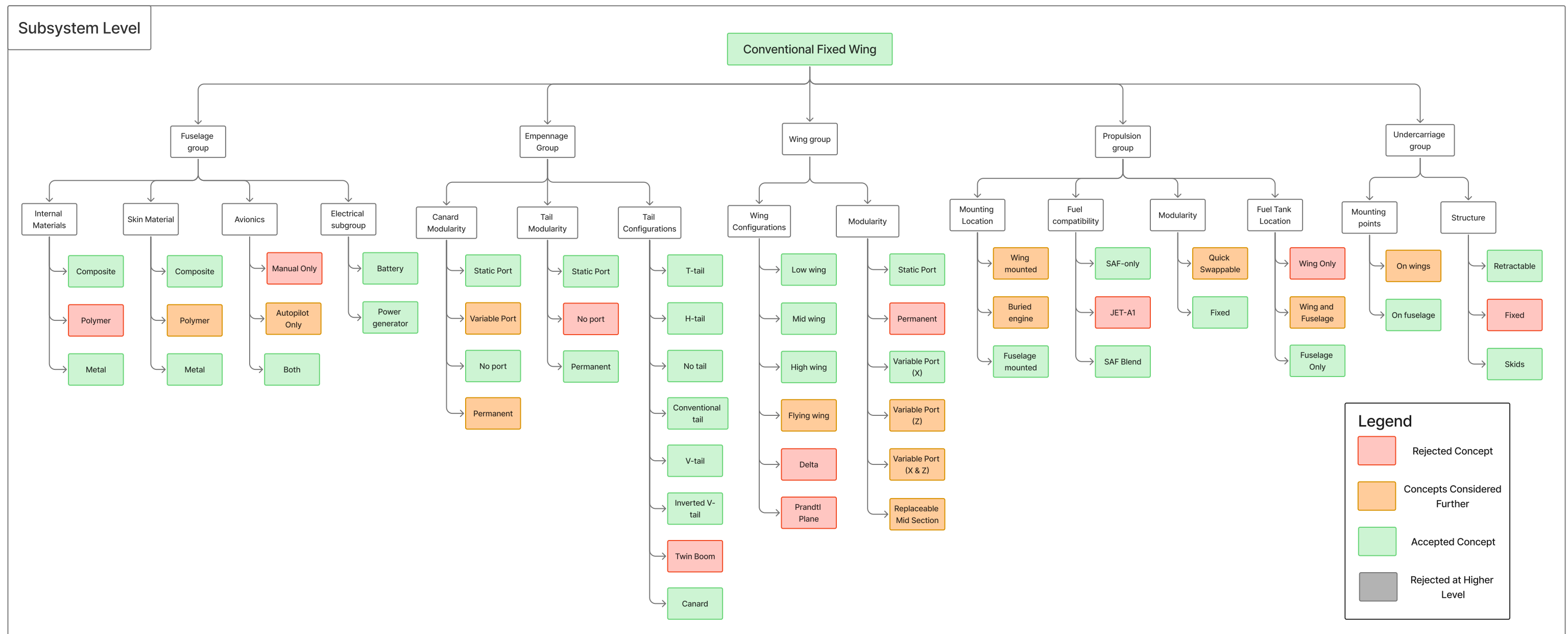
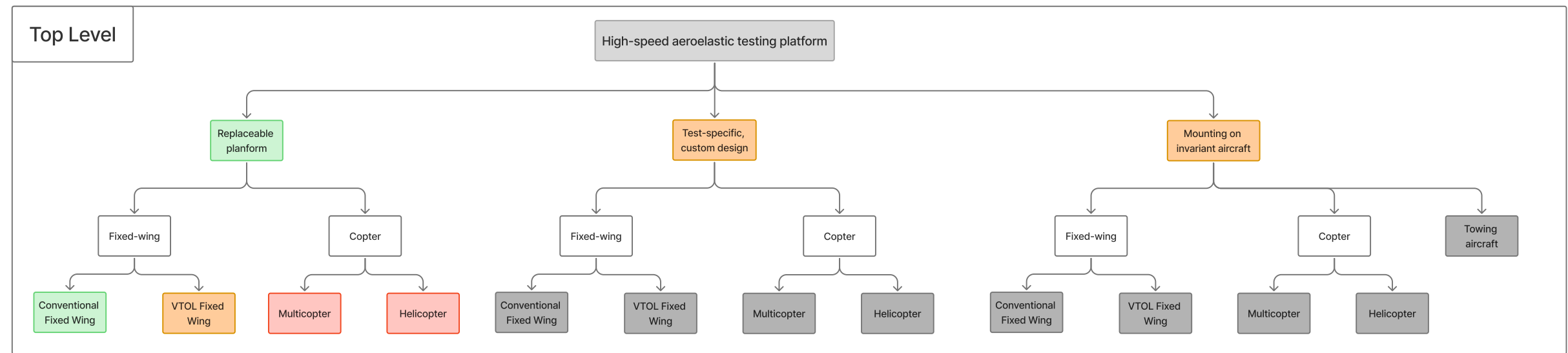


Figure 7.1: Top level and subsystem level Design Option Tree

## 8. Sustainability Analysis

Considerations regarding sustainability have become an important element of modern aerospace engineering. Therefore, in this project, it was very important to design for the sustainable construction and operation of the aircraft. Sustainability was taken into consideration at every step throughout the design process of HUGO.

### 8.1. Construction

Historically, the construction of an aircraft has been a source of large emissions. The requirements for exotic materials with a high degree of processing lead to large energy usage and the generation of waste products. This has to be taken into consideration for the choice of material and design of the aircraft. Emphasis should be placed on reducing material usage by designing structurally efficient components. Another important consideration is that construction is a once-in-a-lifetime emission for the aircraft. This means that a weight saving measure which results in higher production emissions might be worth it for the long-term emission reduction due to weight saving.

### 8.2. Operation

Since it was specified that the aircraft should use the Vulcan turbine engine, it will always generate emissions while in use. It was also specified that this turbine should be powered by SAF (sustainable aviation fuel) meaning it would have a non-petroleum source. This could theoretically lead to net-zero carbon emissions. Though the generation of SAF requires energy and this does not mitigate the emission of NOX typically attributed to turbine jets [22]. It is therefore also relevant to investigate modes of operation and implement systems that would decrease the amount of NOX.

### 8.3. End-Of-Life (EOL) Solutions

The aircraft is designed to have a long lifetime (see: HUG-STR-012) , but it is also important to design for the point when the structure needs to be scrapped. Recycle-ability should be prioritized for the aircraft's components. This is to be considered in the design of the aircraft. The material choice and bonding methods will influence how difficult (or whether it is even possible) to recycle the used material.

### 8.4. Life Cycle Assessment (LCA)

To quantify the material input and output and the subsequent consequences of these the Life Cycle Assessment model is used to evaluate the more detailed concepts in later stages of the design process. Since a full Life Cycle Assessment is very time consuming and requires a complete finalised design it is relevant to use the streamlined Life Cycle Assessment. This is a version of the model which focuses on the differences between two designs and can be used to quickly evaluate more different designs[25].

### 8.5. Environmental Benefits

Another important aspect to consider is the overall environmental impact of the research enabled by the system. An emphasis was placed on how the system could enable a more detailed understanding of the operation of high-aspect ratio wings. The utilization of such technologies could potentially lead to major reductions in aviation-emissions (by some estimates up to 10% [18]). The major benefit of this product is the rapid testing and iteration of potential fuel-saving designs that it enables.

## 9. Profitability and return analysis

To estimate the profitability and return of the HUGO, it is necessary to determine how often the aircraft is going to be utilized. The duration of one flight test is expected to be 30 minutes (as stated in **HUG-OPS-004**) and multiple flight tests can be conducted during a test day. A reference test day was determined to include four test flights. It is assumed that the UAV will be actively operational 100 days a year. This is determined from all working days in a year with a 20% downtime due to maintenance, 60% of days available with favourable weather conditions, and 80% usage of remaining days. Since those assumptions are arbitrary, it is crucial to include the number of flights a year as a sensitivity study parameter.

### 9.1. Market price

It has been decided that the UAV will be made available to customers by leasing. A customer will pay a fee determined by the number of flight days, analogous to a typical wind tunnel financial model. In the price of a typical flight day, all consumables and operations costs are included. These costs are estimated at €450 per typical flight day. Additionally, a fraction of the UAV production, insurance, certification and total 15-year maintenance price is included to return the one-time and per aircraft expenses in the period of 3 years, i.e. 300 flight days. Maintenance cost fractions are based on the expected usage over 15 years, i.e. 1500 flight days with 2 flight hours each. Based on the derived costs, 292 € of profit is needed to break even in 3 years. Overview of the costs considered can be seen in [Table 9.1](#). A detailed description of the costs is present in the subsequent sections.

### 9.2. Market share and volume

To calculate the market share and volume, three main markets were considered: Total Addressable Market (TAM), Serviceable Addressable Market (SAM), Serviceable Obtainable Market (SOM). The total global market for aeronautical testing is reported at \$6.94B in 2026, with a compounded annual growth rate of 7.28% until 2035, representing the TAM <sup>(15)</sup>. A more accessible early market is the European aerospace wind tunnel testing market, reported at \$ 1B <sup>(16)</sup>.

The Netherlands was chosen as the beachhead market, representing the SAM. Public data for the aerospace component testing market was not available. Consequently, two approaches were used to estimate the total relevant Dutch market size. First, a top-down approach starting with the entire Dutch economy, and narrowing to aviation, and aerospace engineering R&D, and finally wind tunnel testing <sup>(17)</sup>. However, this methodology could not provide reasonable estimates relative to the TAM. The bottom-up approach proved much more successful. First, by finding the largest players in the Aerospace component testing industry in the Netherlands, and approximating their total budget for testing on wind tunnels and aircraft like the PH-LAB. The NLR, TU Delft, and DNW were identified as the major players. The TU Delft's total yearly operating costs of the nine wind tunnel facilities were approximated using publicly available data from NASA's wind tunnels in 2012, and adjusting for inflation, as the TU Delft does not report its own operating costs <sup>(18)(19)(20)</sup>. Then, the total operating cost of the PH-LAB was calculated by extrapolating the total flight time of the last three months from publicly available flight logs, and comparing with standard operating costs of the Cessna Citation 550 <sup>(21)(22)</sup>. The NLR also does not make operational costs freely available. Thus, the operational cost breakdown of the

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<sup>15</sup> [Aeronautical testing market](#)(accessed on 24.04.2026)

<sup>16</sup> [European Aerospace wind tunnel testing market](#)(accessed on 24.04.2026)

<sup>17</sup> [Top-Down](#)(accessed on 24.04.2026)

<sup>18</sup> [TU Delft wind tunnel facilities](#)(accessed on 24.04.2026)

<sup>19</sup> [NASA Wind Tunnels](#)(accessed on 24.04.2026)

<sup>20</sup> [Inflation Calculator](#)(accessed on 24.04.2026)

<sup>21</sup> [PH-LAB flight history](#)(accessed on 24.04.2026)

<sup>22</sup> [Cessna Citation 550 Operational costs](#)(accessed on 24.04.2026)

DLR was used, and applied to the NLR <sup>(23)</sup>(24). DNW's funding and operating costs are also not available, so an estimate for their budget was found, and the share of wind tunnels in the Netherlands was used to proportionally divide the testing budget <sup>(25)</sup>(26). These were then summed resulting in a SAM of approximately €71M <sup>(27)</sup>.

Realistically, not all experiments can be performed on the HUGO platform. Estimating 20% of the SAM can be replaced by HUGO, the SOM is thus €14.2M in 2026.

Several assumptions were made regarding the operational costs and total aerospace testing budgets used in the SAM. However, due to estimates such as using the operational costs instead of client test costs, the final SAM and SOM represent a conservative estimate, and are likely to be larger than reported. Nonetheless, key assumptions such as the percentage of SAM that can be addressed by HUGO need to be checked in a sensitivity study.

### 9.3. Development costs

All development costs used for the profit and market price calculations are shown in [Table 9.1](#)

#### 9.3.1. Labour costs

This aircraft is actively developed by TU Delft students. Therefore, the human labour costs associated with the project development, such as tasks associated with design, testing and manufacturing, are not included.

#### 9.3.2. Certification costs

The UAV will be subject to several airworthiness certification procedures. First, the operator of the aircraft has to be registered in The Netherlands under the RDW authority to get authorized for operating UAVs. The cost of this administrative action is approximately €50 <sup>(28)</sup>. Flying a UAV in the specific category under EASA regulation requires permit from the ILT. The fee for this procedure usually lies below €10,000 <sup>(29)</sup>. A cost of €5,000 is estimated here. That being said, since an aircraft of the considered speed requirement and versatility is the first of its kind, certification remains uncertain and certification cost shall be subject to sensitivity study.

### 9.4. Production costs

The total production cost of the aircraft is heavily dependent on the materials and resources used. The maximum production cost per unit is capped at €45,000 [2]. Among the direct costs included in this number are those of the materials used, of facility use and direct labour costs. An initial estimate of 300 man-hours at a wage of €50 per hour is used. The current manufacturing cost estimates allocate approx. 62 % of the production costs to the material and 38% to labour and facility or other associated costs. These figures are analogous to operating expenses in drone manufacturing published by Imarc <sup>(30)</sup>. The UAV production process is estimated to last two working weeks (e.g. 10 working days) with a total facility use expense of €2,000 (i.e. €200 per day). Thus, €28,000 remains for materials/COTS components needed for each aircraft.

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<sup>23</sup>[DLR Budget](#)(accessed on 24.04.2026)

<sup>24</sup>[NLR Budget](#)(accessed on 24.04.2026)

<sup>25</sup>[DNW Budget](#)(accessed on 24.04.2026)

<sup>26</sup>[DNW Wind Tunnels](#)(accessed on 24.04.2026)

<sup>27</sup>[GitHub DSE-28 Financial Plan](#)(accessed on 28.04.2026)

<sup>28</sup>[Port Rotterdam Regulations](#)(accessed on 24.04.2026)

<sup>29</sup>[Applying for a pilot licence or operator number for a drone | RDW](#)(accessed on 24.04.2026)

<sup>30</sup>[Drone Cost Model: From Design to Deployment](#) (Accessed 30.04.2026)

## 9.5. Direct operational costs

Operations of the UAV include several categories of expenses, including fuel and energy costs, transport costs, insurance costs and labour costs.

### 9.5.1. Fuel and energy costs

In order to perform a 30-minute test flight, around 11.5 litres of SAF need to be used. This is based on the average consumption of one VULCAN engine. Considering a SAF price of €2768 per ton <sup>(31)</sup>, at a density of  $800 \text{ kg} \cdot \text{m}^{-3}$  [21], the price per litre of SAF lies at €2.2 per litre. Therefore, the fuel cost per one test flight can be estimated at €30. Other energy involved in the process includes electricity used for charging the UAV batteries and powering computers and hardware needed to conduct the test. In general, these costs will be low. An extra margin of €20 per test flight is added to the total energy cost which accounts for a sufficient margin.

### 9.5.2. Transport costs

Transport costs represent expenses associated with transporting the aircraft to the test site. This includes fuel or energy used by the transport vehicle and the potential cost of leasing a vehicle or hiring an external driver. Assuming that the aircraft is mostly going to be used for testing within The Netherlands, these costs can be estimated at €250 per test.

### 9.5.3. Insurance costs

Since the aircraft will have a maximum take-off mass larger than 20 kg, legislation requires arranging a third-party insurance. The annual price of insurance for a similar-sized UAV usually ranges from lower to higher hundreds of Euros <sup>(32)</sup>. Therefore, an estimate of an annual price of 500 € has been used.

### 9.5.4. Maintenance costs

During its operational life, the UAV will need regular maintenance to remain operational. First, the VULCAN engine should be regularly maintained. For turbines of similar size, such as the Hybl H20PRO, the maintenance interval is around 50 hours <sup>(33)</sup>. This includes inspection and occasional part replacement (e.g. bearings or spark plugs). Therefore, €250 for every 50 operational hours of the VULCAN engine are included for maintenance costs. The maintenance of the airframe is heavily dependent on the use case. It is assumed that the airframe will not require serious maintenance over its whole operational lifetime. A fixed expense of €500 for every 100 flight hours is accounted for to allow for inspection, battery replacement and minor repairs. The effect of maintenance cost also needs to be addressed in a sensitivity study. This study can be conducted once more detail on the trade-off options is known so that any maintainability differences between the options can be captured.

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<sup>31</sup> [EASA](#)(accessed on 24.04.2026)

<sup>32</sup> [DroneLand](#)(accessed on 24.04.2026)

<sup>33</sup> [H20PRO](#)(accessed on 24.04.2026)

**Table 9.1:** Overview of UAV costs and their contribution to flight day pricing

| Type                                                    | Subcategory  | Description                          | Pricing Role | Cost (€) |
|---------------------------------------------------------|--------------|--------------------------------------|--------------|----------|
| <b>Direct costs, reference flight day</b>               |              |                                      |              |          |
| Operations                                              | Turbine fuel | 11.5 L of SAF, 4 test flights        | Direct       | 120      |
| Operations                                              | Electricity  | Batteries, computers, systems        | Direct       | 80       |
| Operations                                              | Transport    | Vehicle + logistics                  | Direct       | 250      |
| Operations                                              | Mission      | UAV pilot/operator                   | Direct       | 400      |
| <b>Fractional costs, full return in 300 flight days</b> |              |                                      |              |          |
| Certification                                           | RDW + ILT    | Registration and flight permits      | Fraction     | 5 000    |
| Production                                              | Materials    | UAV structural and system components | Fraction     | 28 000   |
| Production                                              | Labour       | 300 h at 50 €/h                      | Fraction     | 15 000   |
| Production                                              | Facility     | 10 days at 200 €/day                 | Fraction     | 2 000    |
| Operations                                              | Insurance    | 500 € annually, 15 years total       | Fraction     | 7500     |
| Maintenance                                             | Engine       | 250 € per 50 h, 3000 h total         | Fraction     | 15000    |
| Maintenance                                             | Airframe     | 500 € per 100 h, 3000 h total        | Fraction     | 15000    |

## 10. Technical Risk Assessment

According to EASA risk-based classification rules<sup>(34)</sup>, the HUGO falls under the "specific" category of UAS (Unmanned Aircraft Systems). This classification follows from the design MTOM exceeding 25kg, maximum operational velocity of Mach 0.75, maximum altitude exceeding 120m, and BVLOS (Beyond Visual Line of Sight). To ensure operational safety, a preliminary risk assessment has to be performed for both development and operational risks. A qualitative assessment of risks is performed in [section 10.1](#). Mitigation measures are identified and quantified in [section 10.2](#). Lastly, a preliminary Specific Operation Risk Assessment (SORA) methodology is followed in [section 10.3](#) to determine the necessary measures required to ensure safe operations of HUGO.

### 10.1. Initial Risk Assessment

In risk assessment, key risks are identified by evaluating the likelihood of occurrence  $p$  and severity of impact [26]. For the purpose of this analysis, both of these criteria are evaluated on a scale from 1 to 5, with 1 being the lowest score and 5 being the highest. Details regarding the allocation of both scores are explained in [Table 10.1](#).

**Table 10.1:** *Likelihood/Severity Scores used in Risk Assessment* [27]

| Score | Likelihood         | Severity                                                                                                                                          |
|-------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | $0\% < p < 1\%$    | Negligible                                                                                                                                        |
| 2     | $1\% < p < 30\%$   | Marginal (insignificant impact, can be managed by applying standard procedures)                                                                   |
| 3     | $30\% < p < 50\%$  | Considerable (significant impact, can be managed by additional/contingency procedures)                                                            |
| 4     | $50\% < p < 70\%$  | Critical (possible injury or minor damage to property or HUGO, has to be addressed by contingency/emergency procedures)                           |
| 5     | $70\% < p < 100\%$ | Catastrophic (possible death or serious injury, severe damage to property, loss of HUGO, has to be addressed by contingency/emergency procedures) |

#### 10.1.1. Development Risks

For any engineering project it is vital to identify the risks that exist related to the development process [28]. Finding the development risks for a project requires clearly defined requirements that this system must meet. Then identify everything that might prevent the final designed system from meeting these requirements. These risks can be defined by their severity (the risk they pose for the product to meet the requirements) and the probability of this occurring. The development risks identified for HUGO, along with their assigned likelihood and severity scores, are listed in [Table 10.2](#).

<sup>34</sup>[EASA Specific Category — Civil Drones](#) (accessed on 23.04.2026)

**Table 10.2:** *Initial Development Risk Assessment*

| Risk ID    | Risk Description                                                                                                      | Likelihood | Severity |
|------------|-----------------------------------------------------------------------------------------------------------------------|------------|----------|
| HUG-DEV-01 | Discrepancies between the values used for the dimensions, weight and performance of components and the actual values. | 4          | 4        |
| HUG-DEV-02 | Exceeding the time budget.                                                                                            | 2          | 3        |
| HUG-DEV-03 | System Model Errors (inadequate Validation & Verification).                                                           | 3          | 5        |
| HUG-DEV-04 | Inconsistent documentation (errors in keeping of records, documents and calculations)                                 | 2          | 4        |
| HUG-DEV-05 | Supply Chain issues (component shipping delay)                                                                        | 2          | 4        |

### 10.1.2. Operation risks

Operation risks refer to hazardous scenarios which can occur both on the ground and in the air. They are identified and classified based on the following criteria [29]:

- The extent to which third parties or property on the ground could be endangered by the activity;
- The complexity, performance and operational characteristics of the unmanned aircraft involved;
- The purpose of the flight, UAS type, probability of collision with other aircraft, class of airspace;
- The type, scale, and complexity of the UAS operation or activity, including, where relevant, the size and type of the traffic handled by the responsible organisation or person;
- The extent to which the persons affected by the risks involved in the UAS operation are able to assess and exercise control over those risks.

Table 10.3 lists the operation risks identified for HUGO, along with their respective assigned likelihood and severity scores [30].

**Table 10.3:** *Initial Operation Risk Assessment*

| Risk ID   | Risk Description                                                            | Likelihood | Severity |
|-----------|-----------------------------------------------------------------------------|------------|----------|
| HUG-OP-01 | Loss of control during flight while connection with the drone is maintained | 2          | 4        |
| HUG-OP-02 | Loss of communication with the drone during the test phase                  | 2          | 4        |
| HUG-OP-03 | Human Error                                                                 | 3          | 3        |
| HUG-OP-04 | Unscheduled sudden disassembly during flight                                | 2          | 5        |
| HUG-OP-05 | Exceeding the flight envelope                                               | 2          | 3        |
| HUG-OP-06 | Exceeding the flight limit area                                             | 2          | 5        |
| HUG-OP-07 | Air-collisions                                                              | 1          | 5        |
| HUG-OP-08 | Fuel-fire                                                                   | 2          | 4        |
| HUG-OP-09 | Mechanical failure of components                                            | 3          | 4        |
| HUG-OP-10 | Rapid deterioration of weather conditions during flight                     | 2          | 3        |

### 10.1.3. Initial Risk Map

The development and operation risks identified earlier in this Section can be visually represented by means of a risk map in [Figure 10.1](#) <sup>(35)</sup>. Ideally, most risks shall be located in the green zone of the map, which represents a sufficiently low likelihood of occurrence and severity of impact to not warrant significant concern. However, this is not the case for the initial risk assessment.

Risks located in the yellow zone of the map present significant hazards which can be addressed by means of abnormal/emergency checklists and procedures. The risks located in the red zone of the map are deemed as critical. It is necessary to lower their likelihood and/or severity by developing extensive preventive measures and contingency strategies.

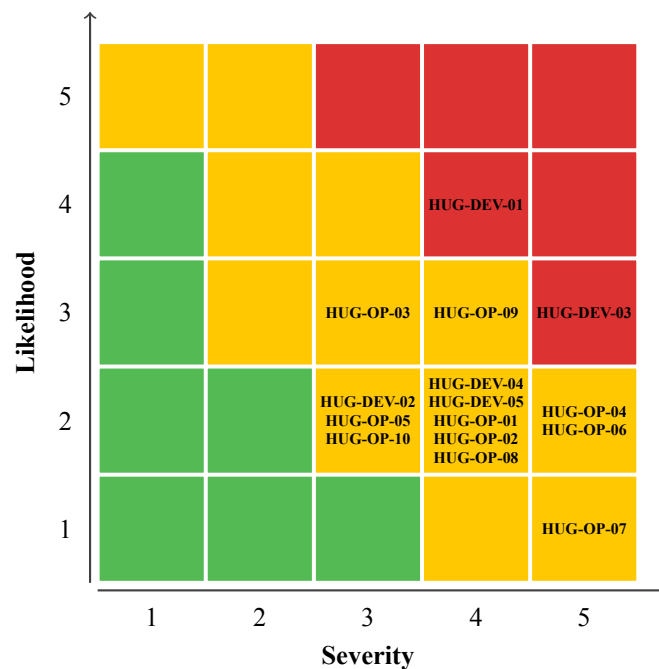


Figure 10.1: Initial Risk Matrix

## 10.2. Contingency Management

Having identified the risks posed to the development of the system, it is possible to explore adequate preventive measures and mitigation strategies. Preventive measures are identified as processes that decrease the likelihood of a risk occurring in the first place, while mitigation strategies are expected to lower the severity of consequences in case a risk manifests during development or operations [27].

[Table 10.4](#) lists the envisioned preventive measures addressing both development and operational risks, as well as their expected effect on the likelihood score,  $\Delta L$ . [Table 10.9](#) lists the envisioned contingency scenarios, along with the expected effect on the severity score,  $\Delta S$ .

<sup>35</sup> [What is a risk map \(risk heat map\)?](#) (accessed on 24.04.2026)

**Table 10.4:** *Risk Preventive Measures*

| <b>Risk ID</b> | <b>Preventive Measure(s)</b>                                                                                                                                                        | <b><math>\Delta L</math></b> |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| HUG-DEV-01     | Engineering margins can be used to ensure the behaviour of the physical system is in line with the predictions made for the theoretical.                                            | -2                           |
| HUG-DEV-02     | Adapting adequate Project Management and Project Planning strategies to minimise avoidable delays due to poor organisation.                                                         | -1                           |
| HUG-DEV-03     | Extensive V&V applied, accounting for model uncertainty and errors.                                                                                                                 | -2                           |
| HUG-DEV-04     | Continuous back-up and synchronisation across platforms, clear organisational structure.                                                                                            | -1                           |
| HUG-DEV-05     | Formulation of detailed manufacturing/production plans for HUGO. Detailed analysis of production chain for vital components, identification of back-up manufacturers and suppliers. | -1                           |
| HUG-OP-03      | Adequate theoretical and practical training of crew. Development of procedures, checklists, flight tests and simulations validated by a competent third party.                      | -1                           |
| HUG-OP-04      | Short service cycles and monitoring of the condition of critical components.                                                                                                        | -1                           |
| HUG-OP-09      | Short service cycles and monitoring of the condition of critical components.                                                                                                        | -1                           |

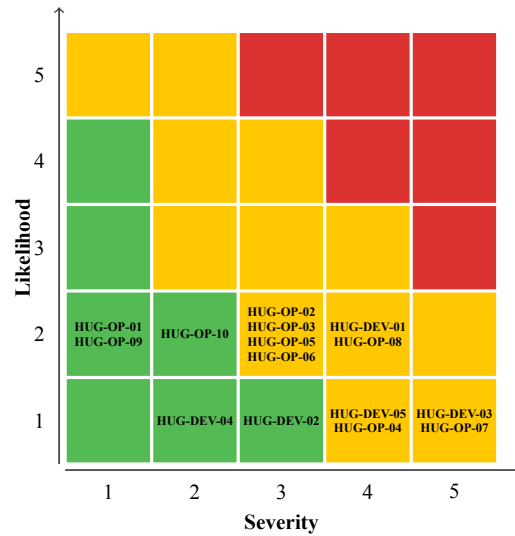
**Table 10.5:** *Risk Mitigation Strategies*

| <b>Risk ID</b> | <b>Mitigation Strategy</b>                                                                                                                                                                                                                                                                                                                                                                                                                               | <b><math>\Delta S</math></b> |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| HUG-OP-01      | For any loss of connection or control that happens for a duration longer than 10 s, a flight termination system shall be triggered.                                                                                                                                                                                                                                                                                                                      | -1                           |
| HUG-OP-02      |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                              |
| HUG-OP-04      | For problems during flight that include breaking of parts or components, HUGO shall immediately decelerate to a safe speed (TBD). Subsequently, a recovery of the vehicle shall be attempted, with the same procedures applicable as for HUG-OP-01.                                                                                                                                                                                                      | -1                           |
| HUG-OP-06      | If the test area is exceeded, the drone will be ordered to immediately return to the testing location.                                                                                                                                                                                                                                                                                                                                                   | -2                           |
| HUG-OP-09      | In the case of a mechanical error onboard the drone during the test flight is detected, an assessment of the extent of the issue shall be performed. Depending on the root cause of the error and the severity of potential consequences, one of the following shall be implemented: <ol style="list-style-type: none"> <li>1. The drone will be ordered to immediately return.</li> <li>2. The flight termination system shall be triggered.</li> </ol> | -1                           |
| HUG-OP-10      | In case of deteriorating weather conditions, the drone will be ordered to return and land as soon as possible.                                                                                                                                                                                                                                                                                                                                           | -1                           |

The final, revised likelihood and severity scores for both development and operational risks are presented in [Figure 10.6](#). The risk matrix of mitigated risks is shown in [Figure 10.2](#).

**Table 10.6:** *Final Development and Operational Risk Assessment*

| Risk-ID    | Likelihood | Severity |
|------------|------------|----------|
| HUG-DEV-01 | 2          | 4        |
| HUG-DEV-02 | 1          | 3        |
| HUG-DEV-03 | 1          | 5        |
| HUG-DEV-04 | 1          | 2        |
| HUG-DEV-05 | 1          | 4        |
| HUG-OP-01  | 2          | 1        |
| HUG-OP-02  | 2          | 3        |
| HUG-OP-03  | 2          | 3        |
| HUG-OP-04  | 1          | 4        |
| HUG-OP-05  | 2          | 3        |
| HUG-OP-06  | 2          | 3        |
| HUG-OP-07  | 1          | 5        |
| HUG-OP-08  | 2          | 4        |
| HUG-OP-09  | 2          | 1        |
| HUG-OP-10  | 2          | 2        |

**Figure 10.2:** *Final Risk Matrix*

As can be seen in Figure 10.2, applying preventive measures and contingency scenarios to the design and operations allows for a significant decrease in risk severity for all cases considered. Notably, none of the risks remains in the red area of the matrix, and numerous options have transitioned into the green zone. Although some options still remain in the yellow due to their high severity, it should be noted that likelihood values were minimised.

### 10.3. Preliminary Specific Operation Risk Assessment (SORA)

The SORA is a classification method used by JARUS (Joint Authorities for Rulemaking on Unmanned Systems) to determine the risk posed by a UAS during operations [29]. Its main purpose is to determine whether the operations can be performed in a safe manner. The nominal SORA method is typically comprised of 10 steps encompassing the ground risk, air risk, mitigation performance requirements, assurance and integrity level estimation, and determination of containment requirements.

In this Section, a preliminary SORA classification will be performed for HUGO based on currently available data. It is expected that the final results of this methodology may change as more information becomes known about the UAS configuration and the mission area.

#### 10.3.1. Documentation of Proposed Operations

The first step in the SORA methodology focuses on compiling operational, technical, and organisational information regarding UAS operations. This entails detailing the operational volume, ground risk buffers, adjacent area, and adjacent airspace [29][31].

At this point in the design phase, a specific operations area has not yet been determined. Based on the available data of HUGO's design (MTOW of 50kg, maximum speed above 200 m/s), a controlled airspace is required with sufficiently low population in the adjacent area [29], which may not be attainable in the Netherlands. For the remaining steps of the SORA, it is assumed that HUGO will operate in controlled areas only. This assumption will impact the determination of ground risk, assurance and integrity levels, and operational objectives, and the results of this analysis should only be considered as a preliminary estimation.

### 10.3.2. Intrinsic Ground Risk Class (iGRC)

The intrinsic Ground Risk Class is determined by the characteristics of the UAS based on its maximum characteristic dimension (wing span), maximum speed, as well as the population density in the operational volume and risk buffer areas [29]. It is expressed by a number from 1 to 10, with higher values corresponding to higher ground risks.

The maximum speed of HUGO exceeds the limit of 200 m/s specified in the JARUS SORA Guide [29]. This implies that the iGRC has to be estimated according to alternative estimation methods defined in Annex F [32]. One of the available trade-offs allows for an increase in maximum velocity by 40%, provided that the maximum size is reduced by 50%. As the expected wingspan for HUGO falls below 20 m, this trade-off is applicable and can be used to estimate the iGRC. Table 10.7 presents an alternative iGRC classification method applicable for the trade-offs.

**Table 10.7:** Alternative Intrinsic Ground Class Value (adapted from [32])

| Alternative Intrinsic Ground Class Value      |            |        |        |                  |         |         |
|-----------------------------------------------|------------|--------|--------|------------------|---------|---------|
| Max UAS Characteristic Dimension              |            | 0.5 m  | 1.5 m  | 4 m              | 10 m    | 20 m    |
| Maximum Speed                                 |            | 35 m/s | 49 m/s | 105 m/s          | 168 m/s | 280 m/s |
| Max population density [ppl/km <sup>2</sup> ] | Controlled | 1      | 1      | 2                | 3       | 3       |
|                                               | < 5        | 2      | 3      | 4                | 5       | 6       |
|                                               | < 50       | 3      | 4      | 5                | 6       | 7       |
|                                               | < 500      | 4      | 5      | 6                | 7       | 8       |
|                                               | < 5000     | 5      | 6      | 7                | 8       | 9       |
|                                               | < 50,000   | 6      | 7      | 8                | 9       | 10      |
|                                               | > 50,000   | 7      | 8      | Not part of SORA |         |         |

Provided that HUGO only operates within controlled areas, it can be assigned an iGRC of 3 according to the alternative classification method.

### 10.3.3. Final Ground Risk Class (GRC)

If the iGRC value is too high, it can be modified or lowered based on risk mitigation scenarios described in the JARUS SORA Guide [29]. However, an iGRC of 3 is considered low enough for the purpose of classification of HUGO, and therefore the iGRC and GRC both attain a value of 3.

### 10.3.4. Air Risk Class (ARC) and Tactical Mitigation Performance Requirement (TMPR)

The SORA methodology uses the operational airspace of the UAS to determine an intrinsic risk of collision with (manned) aircraft, which is expressed by the ARC [29]. For a UAS operating in atypical airspaces (such as reserved/restricted airspaces or defined drone testing areas), such as HUGO, an ARC-a score is automatically attributed.

The ARC can be lowered or modified via strategic and tactical mitigation means [32]. However, since ARC-a represents the lowest possible ARC score, such steps are not deemed necessary for HUGO.

The TMPR refers to operational requirements which shall be applied to mitigate residual risks of air collisions with (manned) aircraft and achieve an acceptable level of safety during UAS operations [29]. For ARC-a operations, the risk of encounter of manned aircraft is deemed as "extremely low", and therefore no TMPR shall be applied (although it is possible for the CAA to require a demonstration of mitigation procedures for HUGO if deemed necessary)[31][33].

### 10.3.5. Specific Assurance and Integrity Levels (SAIL)

The SAIL is a value ranging from I to VI used for the identification of Operational Safety Objectives (OSOs) related to the robustness (integrity and assurance) level of the UAS [29]. Qualitatively, this metric represents the confidence in the safety of operations, and can be determined according to Table 10.8. Based on the final GRC and residual ARC values derived for HUGO, it can be assigned a SAIL II.

**Table 10.8:** *SAIL Determination Table, adapted from [29]*

| <b>SAIL Determination</b> |                                  |          |          |          |
|---------------------------|----------------------------------|----------|----------|----------|
|                           | <b>Residual ARC</b>              |          |          |          |
| <b>Final GRC</b>          | <b>a</b>                         | <b>b</b> | <b>c</b> | <b>d</b> |
| $\leq 2$                  | I                                | II       | IV       | VI       |
| 3                         | II                               | II       | IV       | VI       |
| 4                         | III                              | III      | IV       | VI       |
| 5                         | IV                               | IV       | IV       | VI       |
| 6                         | V                                | V        | V        | VI       |
| 7                         | VI                               | VI       | VI       | VI       |
| $> 7$                     | Category C (Certified) operation |          |          |          |

### 10.3.6. Determination of Containment Requirements

Containment Requirements are compiled to ensure that both ground and air risk fall within the required safety levels in the adjacent area [29]. Based on the SAIL value and UAS characteristic dimensions and maximum speed, a level of robustness can be estimated, along with a set of corresponding safety requirements. A SAIL II UAS flying at maximum speed above 200 m/s in a controlled airspace with population density in the adjacent area not exceeding 50 ppl/km<sup>2</sup> corresponds to a Medium robustness (integrity and assurance) level [29]. The Containment Requirements based on a Medium level of integrity are listed in Table 10.9.

**Table 10.9:** *Containment Requirements based on Integrity [34]*

| No. | Title                                             | Criterion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Operational Volume Containment                    | No probable single failure of the UAS or any external system supporting the operation shall lead to operation outside of the operation volume OR The probability of the failure condition “UA leaving the operational volume” shall be less than 10 <sup>-3</sup> /Flight Hour (FH).                                                                                                                                                                                                                                                             |
| 2   | End of Flight upon exit of the operational volume | When the UA leaves the operational volume, an immediate end of the flight must be initiated through a combination of procedures/processes and/or available technical means.                                                                                                                                                                                                                                                                                                                                                                      |
| 3   | Definition of the final ground risk buffer        | Ground risk buffer must consider the following points: <ul style="list-style-type: none"> <li>• Probable single failures (including the projection of high-energy parts such as rotors and propellers) which would lead to an operation outside of the operational volume,</li> <li>• Meteorological conditions (e.g., maximum sustained wind),</li> <li>• UAS latencies (e.g., latencies that affect the timely manoeuvrability of the UA),</li> <li>• UA behaviour when activating a technical containment measure, UA performance.</li> </ul> |
| 4   | Ground risk buffer containment                    | <ul style="list-style-type: none"> <li>• No single failure of the UAS or any external system supporting the operation shall lead to operation outside of the ground risk buffer.</li> <li>• Software (SW) and Airborne Electronic Hardware (AEH) whose development error(s) could directly lead to operations outside of the ground risk buffer shall be developed to an industry standard or methodology recognised as adequate by the competent authority.</li> </ul>                                                                          |

To demonstrate that the required level of integrity is achieved, it is necessary to present supporting evidence for some of the criteria listed in [Table 10.9](#). According to Annex E of the JARUS SORA Guide [34]:

- For Criterion 1 and Criterion 4: a design and installation appraisal including at minimum:
  - design and installation features (e.g., independence, separation or redundancy claims);
  - any relevant particular risk (e.g., hail, ice, snow, electro-magnetic interference...) associated with the operation and how they are being addressed.
- For Criterion 2: Adequacy of the Emergency Procedures to terminate flight are proven through:
  - dedicated flight tests, or
  - simulation provided the simulation is proven valid for the intended purpose with positive results.

### 10.3.7. Identification of Operational Safety Objectives (OSO)

In this step of the SORA methodology, the SAIL score of the UAS can be used to determine the required level of robustness for each of the Operational Safety Objectives defined in the JARUS SORA Guide [29]. [Table 10.10](#) lists the Recommended OSOs for HUGO based on the corresponding OSO integrity level. The steps to be taken to demonstrate the required level of assurance for each of the OSOs can be found in Annex E of JARUS SORA Guide [34].

**Table 10.10:** Recommended OSOs for HUGO [29][34]

| No. | OSO                                                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Ensure the Operator is competent and/or proven            | <p>The applicant is knowledgeable of the UAS being used and as a minimum has the following relevant operational procedures: checklists, maintenance, training, responsibilities, and associated duties.</p> <p>The applicant has an organization appropriate for the intended operation, with at least the following in place:</p> <ul style="list-style-type: none"> <li>• a method to continuously evaluate whether the operator is operating according to the terms of the operational authorization and check whether the mitigations proposed as part of the operational authorization are still appropriate;</li> <li>• occurrence analysis procedures and reporting to the designer in case of design-related in-service events.</li> </ul>                                                                                                 |
| 2   | UAS manufactured by competent and/or proven entity        | As a minimum, manufacturing procedures cover: specification of materials, suitability and durability of materials used, processes necessary to allow for repeatability in manufacturing and conformity within acceptable tolerances, configuration control.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 3   | UAS maintained by competent and/or proven entity          | <p>The UAS Operator maintenance instructions and requirements are defined, cover, when applicable, the UAS designer instructions and requirements, and are adhered to.</p> <p>The maintenance staff is competent and has received authorisation to carry out UAS maintenance.</p> <p>Preventive/scheduled maintenance and inspection of each UAS is organized and in accordance with the UAS Operator maintenance programme on the basis of the UAS designer scheduled maintenance requirements, and adapted to the specificities of UAS operations.</p> <p>Upon completion, the maintenance log system is used to record all maintenance conducted on the UAS, including releases. A release to service can only be accomplished by a staff member who has received authorisation to release to service for that particular UAS model/family.</p> |
| 5   | UAS is designed considering system safety and reliability | The equipment, systems, and installations are designed to minimize hazards in the event of a probable failure of the UAS or of any external system supporting the operation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 6   | C3 link characteristics are appropriate for the operation | <p>The applicant determines that performance, RF spectrum usage and environmental conditions for C3 links are adequate to safely conduct the intended operation.</p> <p>The UAS remote pilot has the means to continuously monitor the C3 performance and ensures the performance continues to meet the operational requirements.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 7   | Conformity check of the UAS configuration                 | <p>The Operator has UAS conformity check procedures ensuring periodically that:</p> <ul style="list-style-type: none"> <li>• the UAS intended to be used for the operation is in a condition for safe operation,</li> <li>• the UAS configuration conforms to the UAS design data, including any design limitations, e.g. maximum payload weight, considered under the approved concept of operation.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                   |

*Continued on next page*

| No. | OSO                                                                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8   | Operational procedures are defined, validated and adhered to              | Operational procedures appropriate for the proposed operation are defined and as a minimum cover the following elements: flight planning, pre- and post-flight inspections, procedures to evaluate environmental conditions before and during the mission, including real-time evaluation, assessment of meteorological conditions such as METAR and TAFOR with a simple recording system, procedures to cope with unintended adverse environmental conditions, normal procedures, contingency procedures and emergency procedures.                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 9   | Remote crew trained and current                                           | <p>The theoretical and practical training:</p> <ol style="list-style-type: none"> <li>ensures knowledge of: <ol style="list-style-type: none"> <li>UAS regulation,</li> <li>UAS airspace operating principles,</li> <li>airmanship and aviation safety,</li> <li>human performance limitations,</li> <li>meteorology and assessment of meteorological conditions,</li> <li>navigation and charts,</li> <li>UA knowledge,</li> <li>operational procedures and ERP,</li> <li>use of external services, including service limitations and system recovery if applicable.</li> </ol> </li> <li>is adequate for the operation, i.e. allows the remote crew to control normal, abnormal and emergency situations potentially resulting from technical issues with the UAS or external systems supporting UAS operation, human errors or critical environmental conditions.</li> <li>specifies proficiency requirements and training recurrence.</li> </ol> |
| 13  | External services supporting UAS operations are adequate to the operation | <p>The applicant ensures that the level of performance for any externally provided service critical for the safety of the flight is adequate for the intended operation. If the externally provided service requires communication between the Operator and the Service Provider, the applicant ensures there is effective communication to support the service provisions.</p> <p>Roles and responsibilities between the applicant and the external Service Provider are defined.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 16  | Multi crew coordination                                                   | <p>Procedure(s) to ensure coordination between the crew members and robust and effective communication channels is (are) available and at a minimum cover:</p> <ul style="list-style-type: none"> <li>assignment of tasks to the crew, and</li> <li>establishment of step-by-step communications, including a proper phraseology between the remote crew members involved in the aerial part of the operation.</li> </ul> <p>Remote Crew training covers multi-crew coordination. In addition, the Remote Crew receives Crew Resource Management (CRM)3 training.</p>                                                                                                                                                                                                                                                                                                                                                                                |
| 17  | Remote crew is fit to operate                                             | <p>The applicant has a policy defining the criteria and the means for the remote crew to declare themselves fit before starting their duty and report themselves unfit, if required, during their shift. In addition:</p> <ul style="list-style-type: none"> <li>Duty, flight duty and resting times for the remote crew are defined by the applicant and adequate for the operation.</li> <li>The Operator defines requirements appropriate for the remote crew to operate the UAS.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

*Continued on next page*

| No. | OSO                                                                                          | Description                                                                                                                                                                                                                                                          |
|-----|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 18  | Automatic protection of the flight envelope from human errors                                | The UAS incorporates automatic protection of the flight envelope to prevent the remote pilot from making any single input under normal operating conditions that would cause the UA to exceed its flight envelope or prevent it from recovering in a timely fashion. |
| 19  | Safe recovery from human error                                                               | Systems detecting and/or recovering from human errors are developed to industry best practices.                                                                                                                                                                      |
| 20  | A Human Factors evaluation has been performed and the HMI found appropriate for the mission. | The UAS information and control interfaces are clearly and succinctly presented and do not confuse, cause unreasonable fatigue, or contribute to remote crew error that could adversely affect the safety of the operation.                                          |
| 23  | Environmental conditions for safe operations defined, measurable and adhered to              | Environmental conditions for safe operations are defined and reflected in the flight manual or equivalent document.                                                                                                                                                  |
| 24  | UAS designed and qualified for adverse environmental conditions                              | The UAS is designed to perform as intended in the environmental conditions defined and reflected in the flight manual or equivalent document.                                                                                                                        |

#### 10.3.8. Comprehensive Safety Portfolio (CSP)

The Comprehensive Safety Portfolio (CSP) compiles all elements of the SORA which shall be presented to the competent authority for the application of authorisation for UAS operations [32]. Typically, it includes the finalized detailed operational description, all safety claims and their robustness (iGRC and final GRC, initial and residual ARC, TMPR, the OSOs associated with the SAIL, and the containment requirements), compliance evidence, necessary linkages and references between documents, and a finalised compliance matrix.

At the current development stage of HUGO, it is not yet possible to compile a complete version of the CSP.

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